# Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

Anonymous submission

## Abstract

When political content spreads on social media, does diffusion follow universal structural patterns or are these patterns shaped by platform-specific communities? We address this question by comparing how cascades during the 2024 U.S. presidential campaign scale on two ideologically distinct but technically comparable platforms: Truth Social and Bluesky. Using repost and reply networks constructed from public posts mentioning Biden or Trump (May–June 2024), we examine whether similarly sized cascades take similar shapes across platforms. We find that amplification (reposts) scales similarly. Conversational diffusion, however, diverges sharply: reply cascades on Truth Social are wider and shallower, while those on Bluesky are narrower and deeper. We evaluate candidate explanations: influencer dominance, topic variability, ideological composition, and cross-ideological interaction alignment. Neither influencers nor topic differences account for this divergence. Only when both ideological composition and interaction alignment are modeled together do these platform-level scaling differences largely attenuate. Fine-grained motif analysis further reveals asymmetric interaction patterns, including distinct roles for left-, right-, and center-leaning users across platforms. Overall, the findings show that platform differences matter most where political diffusion becomes conversational. Our paper highlights that next-generation social media platforms do not simply transmit political information at different volumes; they organize different opportunities for political engagement and disagreement across ideological groups.

## 1 Introduction

Social media platforms facilitate information cascades, reposts, replies, and other interactions through which political content circulates online (Easley and Kleinberg 2010; Gruhl et al. 2004; Guille et al. 2013). A central question in computational social science is whether the structural patterns of these cascades are governed by universal dynamics or shaped by the particular environments in which they unfold. While there is past work on how diffusion structures vary across information types (e.g., false vs. real news (Vosoughi, Roy, and Aral 2018)), we know little about how diffusion scaling properties vary across platforms.

This question has become increasingly relevant because the social media ecosystem has fractured. After the January 6 Capitol attack, Donald Trump launched Truth Social as a right-leaning alternative to mainstream platforms. More recently, Twitter's rebranding as X and its loosening of moderation have spurred many centrist and left-leaning users to migrate to Bluesky (Gerard, Botzer, and Weninger 2023; Failla and Rossetti 2024; Quelle and Bovet 2025). Although both platforms mimic Twitter's design, follower networks, reposts, and reply mechanics, their contrasting community compositions create distinct social contexts. This creates a natural testbed to determine whether similar technical designs produce similar diffusion structures when embedded in different communities.

To answer this question, we compare the structural diffusion of political content on Truth Social and Bluesky during a high-salience period of the 2024 U.S. presidential campaign. The analysis centers on a one-month window that included major events such as the first presidential debate, focusing on posts containing hashtags and keywords related to "Biden" and "Trump." To characterize diffusion dynamics, we go beyond raw comparisons of cascade size, depth, or breadth. Such comparisons across platforms can be misleading since platforms with more users mechanically generate larger cascades (Juul and Ugander 2021). Instead we focus on how cascades grow as they scale. We compute scaling to determine whether cascades tend to broaden horizontally or deepen vertically as they grow in the number of posts. If scaling relationships are platform-specific, identical levels of engagement produce fundamentally different conversational geometries. Our goal is to understand if such divergence happens and why.

When performing cross-platform comparisons, we also distinguish between diffusion modes. Reposts and replies serve different functions: reposts are acts of amplification that redistribute content to new audiences, while replies are conversational acts that extend, contest, or elaborate on content (Conover et al. 2011; boyd, Golder, and Lotan 2010; Guerrero-Solé 2018). Past work has identified important distinctions between these diffusion modes, such as differences in homophily (Conover et al. 2011). However, cascade research typically aggregates both into general engagement metrics. By treating repost and reply cascades as distinct objects, we can test whether cross-platform universality holds for amplification, for conversation, or for both.

We find that platform differences matter most where political diffusion becomes conversational. Repost cascades scale similarly across Truth Social and Bluesky. Reply cascades, by contrast, diverge sharply: Truth Social produces wider and shallower interactions, while Bluesky produces narrower and deeper conversational threads.

What explains these fundamental differences? Platforms can vary in the dominance of their influential users and this can impact diffusion patterns (Goel et al. 2016; Zannettou et al. 2018). Similarly, certain topics can be more contentious, leading to different diffusion patterns (Vosoughi, Roy, and Aral 2018; Wang et al. 2022) and topic distribution can vary across platforms. Interestingly, we find neither of these explain the reply scaling differences between the two platforms. Ideological composition partially explains the differences and, most importantly, when accounting for both composition and the degree to which users engage in cross-ideological discussions, most of the platform-level difference is accounted for.

Motif analysis helps unpack what this aggregate alignment pattern means locally. It shows that cross-ideological interaction takes different structural forms across platforms: on Truth Social, disagreement more often appears as parallel replies around the same parent post, whereas on Bluesky it more often appears as sequential chains that extend conversations across turns.

These findings matter for understanding the fragmented social media ecosystem. Newer platforms are often built around familiar Twitter-like features, but similar technical designs do not guarantee similar political interaction. Our findings show that political information can be amplified in similar ways while being discussed in very different ways, with replies revealing how platform communities organize opportunities for ideological engagement, disagreement, and visibility.

## 2 Related Work

### 2.1 Comparative Cross-Platform Analysis

A growing body of research argues that social media should be studied as a fragmented ecosystem of platforms rather than as a single, unified public sphere (Rowe and Alani 2014; Boulianne and Larsson 2023). Comparative work shows that political ideology predicts both platform adoption and political use, while platform-specific networking features shape what kinds of political exposure and expression occur there (Boulianne, Hoffmann, and Bossetta 2025). At the ecosystem level, recent scholarship suggests that online fragmentation increasingly takes the form of “echo platforms,” in which ideological differentiation emerges not only within platforms but between them as users migrate across services with distinct moderation regimes, recommendation logics, and community norms (Di Martino et al. 2025; Mosleh, Allen, and Rand 2025).

This perspective is especially important for the study of alternative and emerging platforms. Research on Gab established early that alt-tech platforms can attract ideologically concentrated user bases and sustain different information environments from those observed on mainstream sites (Zannettou et al. 2018). More broadly, Cinelli et al. (2021) show that homophilic clustering and like-minded information diffusion vary substantially across platforms, indicating that similar affordances do not guarantee similar interaction dynamics. Their comparison of Facebook, Twitter, Reddit, and Gab underscores the importance of platform-specific interaction environments in shaping information exposure and segregation.

Work on Truth Social remains comparatively sparse, but existing studies already suggest that it is best understood as a politically concentrated alt-tech environment rather than simply a new campaign tool. Early dataset work positioned Truth Social as a refuge for users alienated by mainstream moderation and documented its rapidly emerging network and content ecology (Gerard, Botzer, and Weninger 2023). Subsequent work released a large 2024 election dataset with posts, replies, and user interactions, emphasizing the platform’s role in hosting contentious, often hyper-partisan political discourse (Shah et al. 2024). More recent network analysis similarly characterizes Truth Social as ideologically homogeneous and structurally segmented, with attention concentrated around a small set of central actors (Hughes and Weninger 2025).

By contrast, recent work on Bluesky describes a Twitter-like platform with a markedly different political center of gravity. Early studies characterized Bluesky as an emerging alternative to X and documented its rapidly expanding data ecology, including large-scale post histories, follow relations, replies, reposts, quotes, and feed-generator interactions (Failla and Rossetti 2024). More recent work shows that Bluesky exhibits heavy-tailed, highly clustered network structure similar to larger platforms, but that users share predominantly left-center news sources and make limited use of custom algorithmic feeds (Quelle and Bovet 2025). Nogara et al. (2026) further find that Bluesky maintains relatively high levels of original posting, low toxicity, and high-credibility news sharing even after rapid growth following its public launch. Together, these studies position Truth Social and Bluesky as a valuable comparative pair: technically comparable platforms embedded in sharply different ideological communities.

### 2.2 Information Diffusion and Cascade Structure

Research on information diffusion has developed a rich vocabulary for describing how online content spreads through networks, including cascade size, depth, breadth, and structural virality (Gruhl et al. 2004; Guille et al. 2013; Easley and Kleinberg 2010; Goel et al. 2016). This literature has linked cascade structure to content attributes such as veracity, novelty, emotional tone, and topic salience. Most prominently, Vosoughi, Roy, and Aral (2018) show that false news on Twitter spreads farther, faster, deeper, and more broadly than true news. Further work examines how topic salience (Wang et al. 2022), cascade predictability (Cheng et al. 2014), and misinformation propagation (Zhao et al. 2020) shape diffusion. Recent work continues to separate these dimensions, distinguishing broadcast-like spread from long peer-to-peer chains (González-Bailón et al. 2024) and depth from breadth when assessing platform interventions (Slaughter et al. 2025). A parallel line asks whether such structures can be measured when platforms do not record diffusion paths: DeVerna et al. (2024) show for Twitter and Bluesky that reconstruction assumptions can distort the resulting cascades.

A related development concerns interaction type. Diffusion studies often prioritize reposting or retweeting because these traces are easy to reconstruct and map directly onto dissemination. Reposting is especially important for political amplification: experimental evidence from Facebook shows that reshared content substantially increases exposure to political news and low-quality sources, demonstrating the centrality of reshare mechanisms to political visibility (Guess et al. 2023a). Foundational work on retweeting also shows that reposts are socially meaningful acts of redistribution, attribution, and audience management, rather than neutral transmission alone (boyd, Golder, and Lotan 2010; Park and Kaye 2019). Who does the resharing also matters: on Twitter/X, a small elite of high-influence accounts carries roughly half of all repost-cascade flow (Niitsuma et al. 2025).

Yet reposting is only one mode of information flow. Prior work shows that retweets, mentions, replies, and quote tweets capture different forms of political engagement. In U.S. election discourse, retweet networks are highly segregated by ideology, whereas mention networks are more politically heterogeneous and contain more cross-ideological interaction (Conover et al. 2011). Studies of political discussions similarly treat retweeting, mentioning, and replying as distinct behavioral traces, with retweet networks more likely to resemble echo chambers and replies or mentions more likely to reflect direct interaction (Guerrero-Solé 2018). Other work finds that emotional and textual features affect replies and retweets differently, suggesting that response-oriented conversation and amplification-oriented redistribution follow different behavioral logics (Kim and Yoo 2012; Toraman et al. 2022). Finally, studies of threaded political conversations show that replies can sustain extended argument across ideological divides, making reply trees especially useful for studying conversational engagement rather than simple visibility (Shugars and Beauchamp 2019).

## 2.3 Ideology, Polarization, and Political Communication Online

A large literature shows that online political communication is shaped by ideological homophily, selective exposure, and asymmetric cross-group engagement (Adamic and Glance 2005; Conover et al. 2011; Flaxman, Goel, and Rao 2016; Del Vicario et al. 2016). Yet this work also suggests that different interaction types produce different political structures. Conover et al. (2011) show that retweet networks are highly segregated, whereas mention networks are more politically heterogeneous. Similarly, Barberá et al. (2015) find that political communication often remains ideologically clustered, but that liberals are more likely than conservatives to engage in cross-ideological dissemination. These studies suggest that exposure to disagreement is unevenly distributed and depends in part on how users interact.

More recent work complicates any simple equation of cross-cutting exposure with deliberative benefit. Bail et al. (2018) show that following opposing political elites on social media can increase polarization rather than reduce it. Garimella et al. (2018) further identify gatekeepers and the "price of bipartisanship," showing that users who bridge polarized communities often occupy structurally costly positions. In parallel, Rathje, Van Bavel, and van der Linden (2021) demonstrate that out-group animosity is a strong driver of sharing, implying that cross-ideological interaction on social media is often conflictual rather than integrative.

Recent evidence also points to ideological asymmetries in what gets diffused. Chang et al. (2025) show that liberals and conservatives differ in the range of issues and the toxicity of elite content they retransmit, suggesting that downstream information environments are shaped not only by whom users follow, but also by what kinds of content each side disproportionately passes along. This raises a related question for cascade analysis: whether political structure operates mainly through user composition, through the alignment of interactions within and across groups, or through both at once.

## 2.4 Theoretical Framework and Hypotheses

**Defining Diffusion Modes and Scaling.** As we mentioned before, we distinguish between two modes of diffusion: reposts and replies. Rather than emphasizing only how far or how fast information travels, our analysis focuses on the shapes of cascades: whether content spreads broadly through direct amplification or develops sequentially through extended exchanges. Because these structural measures all correlate with cascade size (Juul and Ugander 2021), we hold size constant throughout and examine how cascades grow vertically (depth) and horizontally (breadth).

This scaling perspective leads to a broader question: when cascades reach similar sizes, do they take similar structural forms across platforms and diffusion modes, or do they grow through different geometric patterns? If diffusion scaling is broadly universal, similarly sized cascades should exhibit comparable depth and breadth regardless of platform. If scaling is platform-specific, then the same level of participation may produce different cascade shapes depending on the interaction environment.

We therefore examine three candidate explanations for cross-platform variation in cascade geometry. First, user prominence may shape scaling if highly followed or highly active users attract many direct interactions, producing broader cascades. Second, topic variability may matter if some topics invite rapid amplification while others encourage extended discussion. Third, ideological composition and interaction alignment may shape scaling by affecting whether interactions remain within like-minded groups or extend across ideological boundaries.

**H1: Influencer Dominance.** Prior research has demonstrated that a small number of high-follower users can disproportionately influence content dissemination, particularly on platforms characterized by skewed follower distributions (Goel et al. 2016; Zannettou et al. 2018). Truth Social, with its relatively small and ideologically concentrated user base, offers a fertile environment for such dynamics. This is supported by power-law distributions in followers, reposts, and replies per user (see Appendix A), where Truth

Social exhibits a more pronounced heavy-tailed distribution than Bluesky, with a handful of dominant accounts occupying central roles. A clear discontinuity separates the top eight users from the rest, providing a natural cutoff for defining unusually high influence. We hypothesize that this concentration contributes to reply cascades that are broad but shallow, indicative of immediate visibility rather than sustained conversational engagement.

**H2: Ideological Structure.** The ideology of users in a cascade may shape its structure through both participation and interaction. Prior work shows that online political interaction is strongly structured by ideological homophily and selective exposure: amplification networks are often more segregated, while more conversational channels can include greater cross-ideological contact (Conover et al. 2011; Garimella et al. 2018; Barberá et al. 2015; Adamic and Glance 2005; Guess et al. 2023b). However, this literature does not directly establish how ideological alignment affects cascade geometry. We build on it by asking whether ideological composition and cross-ideological interaction are associated with how reply cascades scale in depth and breadth.

We measure ideological structure along two dimensions: (1) *ideological composition*, operationalized as the proportion of posts in a reply tree authored by users in each ideological category, and (2) *interaction alignment*, defined as the share of reply edges connecting users with the same ideological label. Lower alignment indicates more cross-cutting interaction. As shown later in Appendix B, Truth Social and Bluesky differ substantially in the ideological composition of their observed users, making ideological structure a plausible source of cross-platform differences in reply geometry.

We hypothesize that ideological composition and interaction alignment play a significant role in how reply cascades scale. If cross-platform differences in cascade geometry are driven by ideological structure, then incorporating composition and alignment into the models should substantially attenuate platform-level differences in depth and breadth.

**H3: Topic Variability.** Post content shapes how users interact with it. Emotionally charged, controversial, or conspiratorial content often spreads through deeper and more structurally viral cascades than neutral or factual information (Vosoughi, Roy, and Aral 2018; Wang et al. 2022). As described in Section 3.4, we identified eleven salient topics across both platforms; while all are politically related, their distribution differs substantially between Truth Social and Bluesky. We hypothesize that topics play a significant role in how diffusion dynamics scale on social media. For instance, topics involving partisan conflict, institutional distrust, or election-related narratives may elicit distinct structural diffusion patterns depending on their resonance within each platform's community.

**Micro-Level Network Motifs.** To unpack the mechanisms behind platform-level differences in reply cascade geometry, we complement cascade-level regressions with motif analysis, a graph-theoretic approach examining the frequency of small recurring substructures within larger networks (Alon 2007). While statistical models evaluate whether alignment ratio and ideological composition explain divergence, they do not reveal *how* these compositional factors shape micro-dynamics of interaction. We analyze two basic 3-node motif shapes: *chains*, where replies unfold in sequence, and *stars*, where multiple replies attach to the same root. Each motif is labeled by the ideological stance of its participants: Left (L), Center (C), or Right (R).

# 3 Data and Methods

## 3.1 Data Collection

We collected posts and their associated reposts and replies from both platforms during a critical month of the 2024 U.S. presidential campaign, when major political events (including the first debate) spurred intense online discussion. We used each platform's public API to collect posts containing the target keywords and their surrounding conversations. For Bluesky, the thread endpoint enabled retrieval of entire discussion trees around posts mentioning "Biden" or "Trump," while for Truth Social (which follows the Mastodon API structure), we used the context endpoint to capture both ancestor and descendant replies recursively. This approach provides full conversational context rather than isolated messages, allowing us to reconstruct entire reply cascades. For every post and reply, we also collected associated repost actions and each user's follower list. Data were collected between May 30 and June 30, 2024, a high-salience period that included the first U.S. presidential debate.

## 3.2 Cascade Construction

We model information diffusion as directed acyclic graphs (DAGs), where each node is a single post (the root post, a reply, or a repost) and edges represent inferred diffusion pathways, following the time-inferred approach of Vosoughi, Roy, and Aral (2018). Nodes are posts rather than accounts, so a user who replies several times in one thread contributes one node per reply. Ideology, by contrast, is aggregated at the user level: each node carries the label of the account that wrote it (see Ideology Estimation and Alignment), so a cascade's ideological composition is the share of its posts written by left- or right-leaning accounts. For **reply cascades**, trees were constructed directly from reply metadata: each reply is linked to its parent post, and cascades are rooted at the original post with subsequent replies linked recursively. For **repost cascades**, platform APIs report all reposts as nominally linking to the original post, which obscures the true diffusion path. To approximate true repost trees, for each repost $r_i$ we identified all prior reposts $r_j$ with $t_j < t_i$ such that $r_i$ is observed to follow the user who made $r_j$ in the follower snapshot, then inferred the edge $r_j \to r_i$ linking each repost to the most recent eligible prior repost. When no such user existed, the repost was attributed directly to the original post. This procedure infers the most probable diffusion paths based on temporal and social proximity. Because these edges are inferred rather than observed, we refer to repost cascades as reconstructed throughout, and we verify that the repost results are robust to the reconstruction rules examined here (Appendix, Robustness of Repost Cascade Reconstruction).

Reconstruction assumptions can distort cascade properties (DeVerna et al. 2024), so we assess whether the substantive conclusions are robust across several alternative rules. The follower network is a post-collection snapshot without edge creation dates, and was collected at the same time for both platforms.

### 3.3 Cascade Metrics

For each cascade we compute four structural metrics. **Size** ($N$) is the total number of posts in the cascade, including the root. **Breadth** ($B$) is the maximum number of nodes at any single depth level: $B = \max_{d \geq 1} n(d)$, where $n(d)$ is the node count at depth $d$. This captures the widest "layer" of the cascade and differs from root out-degree, which counts only direct children of the root. **Depth** ($D$) is the maximum hop distance from the root to any leaf node: $D = \max_v \operatorname{dist}(\text{root}, v)$. **Structural Virality** (Goel et al. 2016) is the average pairwise shortest-path distance between all nodes in the cascade, distinguishing broadcast-like (low virality) from chain-like viral (high virality) spread. All metrics were log-transformed for modeling due to heavy-tailed distributions.

### 3.4 Topic Modeling

To identify dominant themes and evaluate H3, we applied BERTopic, which integrates contextual sentence embeddings (`all-mpnet-base-v2`), UMAP dimensionality reduction, and HDBSCAN clustering. Topics were extracted using class-based TF-IDF over a preprocessed vocabulary. Two human annotators independently reviewed a sample of representative posts from each cluster to produce consensus-based labels, yielding eleven coherent topic categories (inter-annotator $\kappa = 0.53$; model–Human 1 $\kappa = 0.67$; model–Human 2 $\kappa = 0.67$). Posts not assigned to any cluster were excluded from topic analyses. While both platforms engage heavily with Trump's legal cases and the 2024 election, Bluesky posts are more concentrated around policy and legal accountability, whereas Truth Social emphasizes celebratory or pro-Trump content (see Figure 1).

### 3.5 Ideology Estimation and Alignment

**Post-Level Stance Detection.** We used the `Llama-3.1-8B-Instruct` model to classify each reply into one of five ideological stances (*left*, *lean-left*, *center*, *lean-right*, *right*), providing the original post and preceding replies as context. For robustness, we grouped stances into three categories: *left-leaning* (left, lean-left), *center*, and *right-leaning* (lean-right, right). Annotations were validated against two human annotators on a random sample of 200 replies: inter-annotator agreement was $\kappa = 0.77$ and model–human agreement $\kappa = 0.61$ and 0.74, with class-specific and platform-specific performance and bootstrap intervals reported in Appendix Table 3, together with an analysis showing that the results below are unchanged when labels are randomly corrupted at the measured error rates. Because large language models may encode biases, we treat these labels as probabilistic indicators rather than ground truth.

**User-Level Ideology Inference.** User ideology was inferred by aggregating stance labels across all posts authored by each user. A user was classified as *left-leaning* if more than 60% of their posts were labeled left or lean-left, *right-leaning* if more than 60% were labeled right or lean-right, and *center* otherwise. The center category is therefore a residual: it collects users who reach neither threshold, mixing genuinely moderate users, users who post about politics inconsistently, and users whose stance the classifier could not resolve. Appendix B reports how the results change when composition and alignment are computed over partisan users only. This threshold balances ideological signal with label consistency; aggregate patterns are robust across thresholds from 0.5 to 0.7 (see Appendix B).

**Alignment Ratio.** For each reply edge, we code it as *aligned* if both users share the same ideology and *misaligned* otherwise. The alignment ratio for a cascade is the proportion of aligned reply edges. High ratios indicate ideologically homogeneous discussions; lower ratios reflect cross-cutting exchanges. In this scenario, consider a thread with four replies: a right-leaning user replies to a right-leaning root author, a second right-leaning user replies to the same root, a left-leaning user replies to the root, and that left-leaning comment draws a reply from a right-leaning user. Three of the four edges connect users who share a stance and one crosses stances, so the alignment ratio is $3/4 = 0.75$. The measure is a property of the conversation rather than of any participant: the same user contributes aligned edges in one thread and crossing edges in another.

### 3.6 Statistical Modeling

**Baseline Model.** To rigorously assess how cascade structure varies across platforms, we model the scaling relationship between cascade size and breadth using robust regression with the Huber loss function. This approach mitigates the influence of outliers, common in heavy-tailed cascade data (see Appendix I for diagnostics). Robust estimation bounds the influence of extreme residuals without correcting non-constant error variance, so standard errors come from the sandwich covariance estimator; we use Huber's proposal 2 scale, since the default degenerates here.

Our baseline model examines how cascade breadth and depth scale with size, conditional on platform, including a platform indicator and its interaction with log-transformed cascade size:

$$\begin{aligned}\log(Y_i) = \beta_0 &+ \beta_1 \log(\text{size}_i) \\ &+ \beta_2 \cdot \text{platform}_i \\ &+ \beta_3 \cdot (\log(\text{size}_i) \times \text{platform}_i) + \varepsilon_i, \end{aligned} \tag{1}$$

where $Y_i$ denotes either cascade breadth or depth. The coefficient of interest is $\beta_3$, which captures whether the relationship between cascade size and breadth differs by platform.

To quantify the explanatory contribution of additional covariates, we track percent attenuation of $\beta_3$:

$$\Delta = 1 - \frac{\hat{\beta}_{3,\text{aug}}}{\hat{\beta}_{3,\text{base}}}, \tag{2}$$

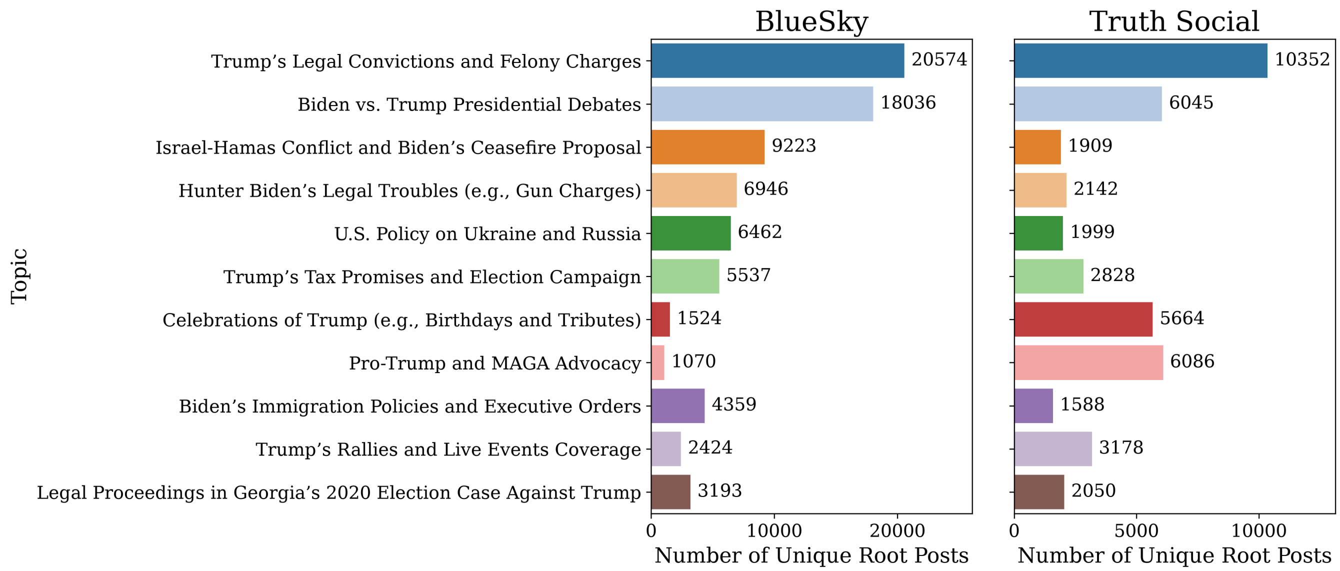


Figure 1: Topic distribution across *Bluesky* and *Truth Social*. Bar plots show the number of unique root posts assigned to each of the 11 topics, based on BERTopic clustering and manual labeling. The panels use independent $x$-scales (Bluesky $n$ = 79,348, Truth Social $n$ = 43,841); compare across panels using the printed counts.

where $\hat{\beta}_{3,\text{base}}$ is the interaction coefficient from Eq. 1 and $\hat{\beta}_{3,\text{aug}}$ is the corresponding estimate from augmented specifications (Models 2–4). A larger $\Delta$ indicates that added covariates account for a greater portion of cross-platform difference.

**Incorporating Influencer Dynamics.** To evaluate H1, we extend the platform variable to three categories: 0 for Bluesky, 1 for Truth Social cascades initiated by non-influencers, and 2 for those initiated by the top eight high-follower users on Truth Social. No comparable concentration of outlier accounts exists on Bluesky.

**Assessing Ideological Composition and Alignment Effects.** Building on user-level ideology labels, we include alignment ratio (spline-transformed, df = 3, to capture non-linear effects at extremes of homogeneity) and the proportions of left- and right-leaning users per reply tree (with a spline for the left-user ratio and a linear term for the right-user ratio, based on partial-residual diagnostics; see Appendix J). Both measures are interacted with cascade size and platform indicators. We decompose this into three specifications: composition only (Model 3a), alignment only (Model 3b), and combined composition and alignment (Model 3c).

**Evaluating Topic Variability.** To address H3, we include cascade-level topic annotations as categorical moderators interacted with $\log(\text{size}_i)$ and platform$_i$, testing whether content differences account for cross-platform variation in how depth and breadth scale with size.

### 3.7 Network Motif Analysis

As introduced above, we use motif analysis to move from cascade-level regression results to the local interaction patterns that compose reply cascade geometry. Here, we describe how motifs are defined, counted, randomized, and scored. We focus on directed 3-node tree motifs, which capture the simplest local distinction between branching and sequential interaction. We distinguish between two topological forms: **star motifs**, where one parent node has two child nodes, and **chain motifs**, where three nodes form a directed path of length two. Each node is annotated with the corresponding user's ideology label: Left, Center, or Right. Motifs are treated as ordered configurations. For chain motifs, order follows the direction of reply; for star motifs, the parent position is distinguished from the child positions. We also summarize ordered motifs into broader families: *aligned* motifs, in which all three users share the same ideology; *one-different* motifs, in which two users share one ideology and the third differs; and *all-different* motifs, in which all three ideological groups appear.

Motifs are counted within each reply tree and then aggregated by platform. For each observed reply tree, the null model preserves the number of nodes and edges while randomizing edge assignments among users within the tree; node ideology labels are held fixed. This preserves the number of cascades, cascade sizes, and platform-specific ideological composition while randomizing local attachment patterns. For each platform, we generate $R = 100$ randomized graphs by double-edge swaps performed within each cascade, with $5m$ swap attempts in a cascade of $m$ edges (Appendix, Motif Frequencies and the Null Model). For motif type $i$, we compute the standardized motif score:

$$Z_i = \frac{N_{\text{obs}}(i) - \mu_{\text{null}}(i)}{\sigma_{\text{null}}(i)},$$

where $N_{\text{obs}}(i)$ is the observed count of motif $i$, and

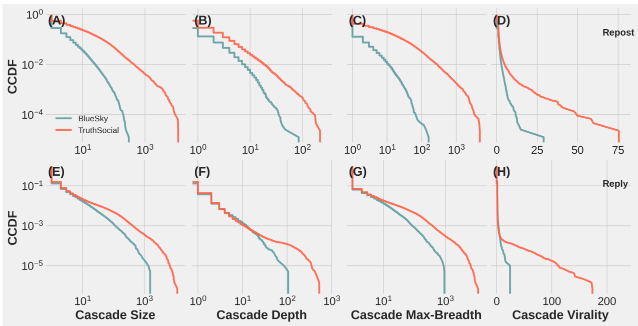


Figure 2: (A–D) Complementary cumulative distribution functions (CCDFs) of repost cascade statistics across two platforms: BlueSky and TruthSocial. Each subplot represents a different structural property: Size, Depth, Breadth, and Structural Virality (Goel et al. 2016). (E–H) The same analyses applied to reply cascades using the same metrics. X-axes are on a log scale (except for structural virality), and y-axes are on a log scale throughout. Each line represents a platform's empirical distribution.

$\mu_{\text{null}}(i)$ and $\sigma_{\text{null}}(i)$ are the mean and standard deviation of its count across randomized realizations. Positive values indicate overrepresentation relative to the null, while negative values indicate underrepresentation. We interpret motif scores by their relative direction and magnitude across platforms rather than treating the motif panel as a set of independent confirmatory hypothesis tests.

## 4 Results

### 4.1 Diffusion Structure Across Emerging Social Networks

As shown in Figure 2, cascades on Truth Social appear larger, deeper, and broader than those on Bluesky for both reposts and replies, and show higher levels of structural virality. A straightforward explanation is platform size, since larger user bases mechanically allow cascades to expand further. Prior work has shown that depth, breadth, and structural virality all correlate positively with cascade size (Juul and Ugander 2021), which raises a concern that marginal comparisons conflate structural effects with sheer volume. To address this, we hold cascade size constant and examine structural differences directly.

Figure 3 presents robust linear fits between log-transformed cascade size and both depth and breadth. After conditioning on cascade size, differences in repost behavior are small: repost cascades scale similarly across Truth Social and Bluesky, with nearly identical slopes. In contrast, reply cascades exhibit a marked divergence. On Truth Social, replies tend to fan out, producing broad but shallow trees, whereas on Bluesky, conversations deepen through sequential exchange, forming narrower but more layered structures. This asymmetry suggests that while baseline amplification behaviors (reposts) are consistent, patterns of dialogic engagement diverge.

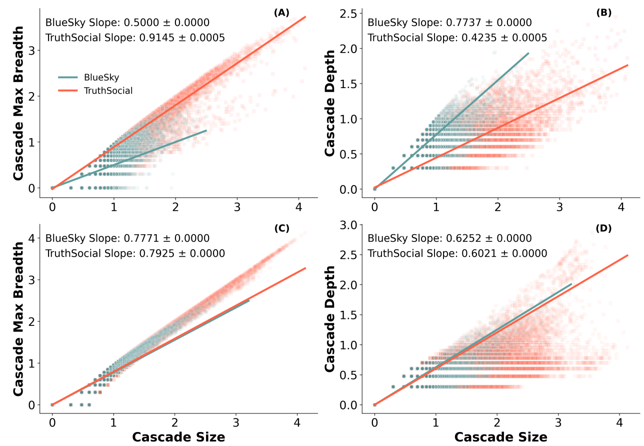


Figure 3: (A–B) Robust Linear Models (RLM) regressions of log-transformed cascade breadth (A) and depth (B) on log-transformed cascade size for reply cascades on BlueSky and TruthSocial. (C–D) Same regressions for repost cascades. Point clouds represent individual cascades (semi-transparent), and lines represent RLM fits.

### 4.2 Mechanisms of Structural Divergence

The modeling results in Table 1 show clear differences in explanatory power across mechanisms. Model 2 (*Influencers*) indicates that cascades initiated by high-follower accounts on Truth Social are only marginally broader, and the platform–size slope gap remains largely unchanged relative to baseline. The broad–shallow structure of Truth Social cascades thus reflects widespread interaction patterns across the platform, not merely the behavior of a small elite.

Model 3 (*Ideological Structure*) is decomposed into three specifications. Composition alone (Model 3a) reduces divergence but leaves substantial residual differences (0.289 for breadth, $-0.196$ for depth). Alignment alone (Model 3b) likewise attenuates but does not eliminate platform differences (0.236 for breadth, $-0.189$ for depth). The combined model (3c) reduces both interaction terms by roughly five-sixths relative to the baseline, to 0.071 for breadth and $-0.049$ for depth against baseline values of 0.419 and $-0.321$, showing that only when both ideological composition and cross-ideological alignment are considered together do platform-specific scaling gaps narrow this far. Model fit confirms this: pseudo-$R^2$ rises to 0.95 for breadth and 0.91 for depth in Model 3c, with corresponding reductions in RMSE. The platform difference is therefore associated with both interaction alignment and broader differences in the ideological makeup of user communities, and neither covariate absorbs it alone. Because alignment is computed from the same reply edges that determine depth and breadth, we read this as statistical accounting rather than as a causal effect; the Limitations section returns to the point.

Model 4 (*Topic Variability*) shows no such effect. The platform-by-size interaction is essentially unchanged from the baseline once topic is included (0.426 against 0.419 for breadth, $-0.330$ against $-0.321$ for depth), indicating that differences in what the two platforms discuss do not account

Table 1: Interaction Effects of Size (Log) and Platform on Breadth (Log) and Depth (Log)

| **Model** | Size (Log) | Size (Log) × TruthSocial | *pseudo* $R^2$ | RMSE |
|---|---|---|---|---|
| **Breadth (Log)** | | | | |
| Model 1 (Baseline) | 0.5043 (0.0010) | +0.4187 (0.0012) | 0.9176 | 0.1319 |
| Model 2 (Influencers) | 0.5042 (0.0010) | +0.4121 (0.0012)† | 0.9182 | 0.1314 |
| Model 3a (Composition) | 0.3692 (0.0014) | +0.2887 (0.0019) | 0.9364 | 0.1159 |
| Model 3b (Alignment) | 0.1713 (0.0012) | +0.2362 (0.0007) | 0.9432 | 0.1095 |
| Model 3c (Combined) | 0.2137 (0.0016) | +0.0706 (0.0015) | 0.9546 | 0.0979 |
| Model 4 (Topic) | 0.5140 (0.0014) | +0.4260 (0.0012) | 0.9191 | 0.1307 |
| **Depth (Log)** | | | | |
| Model 1 (Baseline) | 0.7320 (0.0008) | −0.3214 (0.0009) | 0.8608 | 0.1031 |
| Model 2 (Influencers) | 0.7322 (0.0008) | −0.3134 (0.0009)† | 0.8618 | 0.1027 |
| Model 3a (Composition) | 0.8432 (0.0011) | −0.1955 (0.0015) | 0.8870 | 0.0929 |
| Model 3b (Alignment) | 0.9097 (0.0010) | −0.1889 (0.0006) | 0.8926 | 0.0905 |
| Model 3c (Combined) | 0.9497 (0.0014) | −0.0491 (0.0013) | 0.9085 | 0.0835 |
| Model 4 (Topic) | 0.7168 (0.0011) | −0.3302 (0.0009) | 0.8623 | 0.1025 |

*Notes:* Robust linear models (Huber $M$-estimator) fitted with Huber's proposal 2 scale estimate; asymptotic standard errors in parentheses. Every coefficient shown is more than 25 standard errors from zero, so significance stars are omitted as uninformative. $N = 123{,}189$ for every model.

† Model 2 replaces the binary platform indicator with a three-category variable (Bluesky, Truth Social non-influencer, Truth Social influencer); the entry shown is the Truth Social non-influencer contrast against Bluesky.

for the difference in how their cascades grow.

Taken together, the results clarify the hypotheses. H1 receives limited support: influencer status accounts for only a small share of the platform difference, indicating that Truth Social's broader and shallower reply cascades are not simply produced by high-follower users. H2 receives the strongest support: ideological composition and interaction alignment together account for most of the platform-specific divergence in scaling. H3 is not supported: adding topic leaves the platform-by-size interaction essentially unchanged. These results suggest that conversational diffusion is shaped less by who initiates cascades or what topic they discuss than by the ideological makeup of participants and the alignment of their interactions.

### 4.3 Motif-Level Interaction Patterns

**Chain motifs.** We first examine chain motifs in reply cascades, which capture sequential exchanges where one reply receives another reply. Figure 4 shows how often three-node chain motifs occur relative to randomized reply-tree baselines. We organize these motifs into four ideological configurations: *fully aligned* chains, in which all three users share the same ideological label; *mid-shift* chains, in which the middle user differs ideologically from the first and final users; *end-shift* chains, in which the final user differs from the first two users; and *all-different* chains, in which all three users have different ideological labels. These configurations indicate whether sequential reply structures are built through ideologically aligned exchanges, cross-cutting interventions, or shifts in stance across conversational turns.

Fully aligned Center-labeled chains are overrepresented relative to the null model on both platforms. Fully aligned Right chains fall within $|Z| < 3$ on both platforms: they are common in absolute terms on Truth Social, but no more so than the ideological composition of the platform already implies. Left-leaning chains diverge across platforms: Bluesky shows overrepresentation, whereas Truth Social shows underrepresentation.

The strongest contrasts appear in mid-shift "bounce-back" motifs. On Bluesky, all bounce-back chain motifs are overrepresented, so chains in which the middle user's label differs from the two ends occur more often than the null model predicts, across every ideological pairing. On Truth Social, only Left↔Right bounce-backs are overrepresented, but at much higher magnitudes than on Bluesky. Cross-cutting sequential exchange is therefore broad-based on Bluesky and concentrated on Truth Social, where it appears almost entirely in the Left–Right pairing.

End-shift transitions (e.g., C–C–R, L–L–C) do not follow a uniform pattern. On Bluesky, end-shifts are generally at or below the rewired baseline, with L↔C links most underrepresented. On Truth Social, several center-involved end-shift motifs (R–C–C, R–R–L, L–C–C) are overrepresented, suggesting that the overrepresented cross-cutting configurations involve Center and Left users rather than the Right-leaning majority.

**Star motifs.** We next examine star motifs, which capture local branching in reply cascades: multiple users replying to the same parent post. Whereas chain motifs describe sequential extension across turns, star motifs describe how attention concentrates around a shared conversational object. Figure 5 summarizes standardized motif scores for three-node star motifs in reply cascades. Right-aligned (R–R–R) and center-aligned (C–C–C) stars are strongly overrepresented on Truth Social but suppressed on Bluesky, while left-aligned (L–L–L) stars show $|Z| < 3$ on either platform. This pattern suggests that homophilic clustering is not simply a majority-group phenomenon. Despite being the majority on Bluesky, left-leaning users do not cluster strongly in star formation.

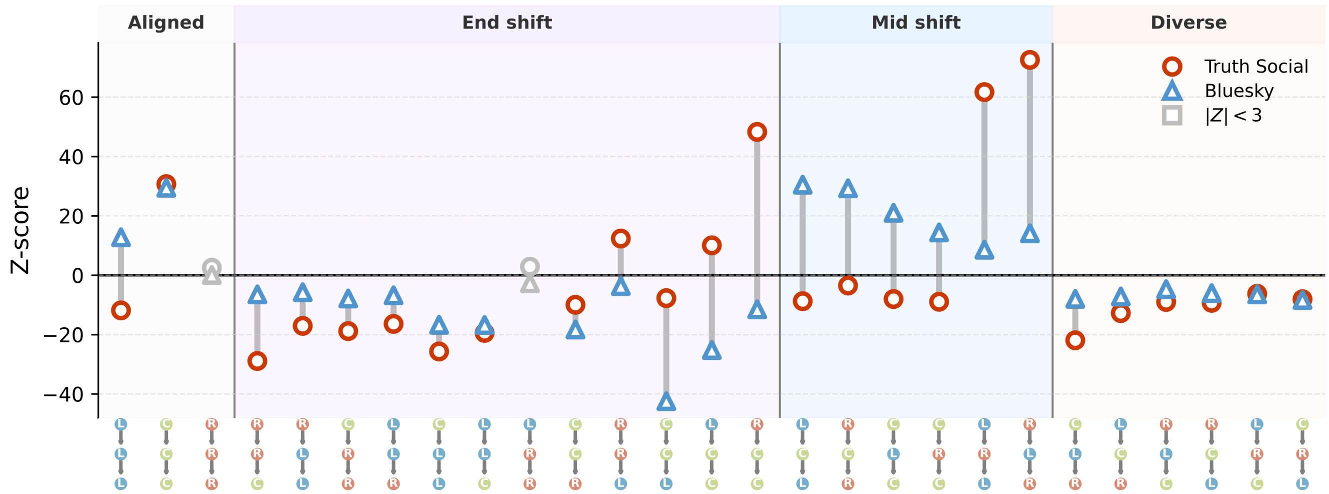


Figure 4: **Chain motif analysis of cross-platform reply structures.** Standardized differences (Z-scores) in the frequency of three-node chain motifs between observed cascades and a randomized null model. Node colors mark ideology (Left = blue, Center = green, Right = red); background shading groups motifs as aligned, mid-shift (middle node differs), end-shift (first or last node differs), or all-different. Circles represent Truth Social, triangles Bluesky; gray symbols denote values with $|Z| < 3$.

Truth Social also shows overrepresentation of several mixed configurations (L–R–R, L–C–C, C–R–R, R–L–L). Most of these patterns correspond to a numerically rare ideology at the root, attracting a more dominant ideology pair as children. These are configurations in which a numerically rare ideology occupies the root and a more common pair occupies the children, which on Truth Social means left-labeled roots with right-labeled repliers.

**Summary.** The motif analysis shows that cross-ideological interaction is not a single structural process. On Truth Social, disagreement more often takes a branching form: multiple users respond to the same parent post, producing local breadth without sustained sequential exchange. On Bluesky, disagreement more often takes a chaining form: replies build on one another, producing depth through multi-turn interaction. Thus, the key difference is not whether ideological difference appears in conversations, but what shape it takes when it appears.

## 5 Discussion

This study compares how candidate-centered conversations about the 2024 U.S. presidential election unfold on two ideologically distinct but technically comparable platforms, Truth Social and Bluesky. Whereas most prior work emphasizes how far or how fast information spreads, we focus on structural form: whether interactions broaden outward through shallow branching or deepen through sequential layering. By analyzing both reposts and replies, we show that reposts scale similarly across platforms, but reply cascades diverge sharply: Truth Social supports broader, shallower exchanges, while Bluesky produces narrower, deeper threads. These results demonstrate that identical affordances can give rise to fundamentally different conversational dynamics, depending on the social and ideological composition of the communities that use them.

Our modeling results show that ideological composition and alignment together statistically account for most of the platform divergence: cross-ideological exchanges are associated with greater depth, and more homogeneous cascades with flatter structures. Yet alignment alone is insufficient. Even when disagreement occurs, reply structures on Truth Social often remain shallow. Motif analysis clarifies this pattern: on Truth Social, disagreement frequently appears as star motifs—parallel replies directed at a single post. On Bluesky, disagreement more often takes the form of chain motifs, where replies build sequentially and support multi-turn interaction.

Motif results highlight the combined role of interactional structure and user composition. On Truth Social, left-leaning users constitute about a tenth of the population, yet appear disproportionately in Left↔Right bounce-back motifs: these are strongly overrepresented relative to the null model while remaining rare in absolute terms. This is why alignment alone cannot account for platform differences: it captures whether interactions cross ideological lines, but not how many users of each ideology are present to cross them.

Center-labeled users appear disproportionately in different local configurations across the two platforms. On Bluesky they occur more often in bounce-back motifs, as the targets of cross-ideological replies; on Truth Social they appear more often in several end-shift configurations. Because the center category is residual, we interpret these patterns as differences involving center-labeled users rather than as evidence about a distinct centrist group.

Although Truth Social and Bluesky provide similar affordances for posting, reposting, and replying, we do not interpret the observed differences as direct effects of platform

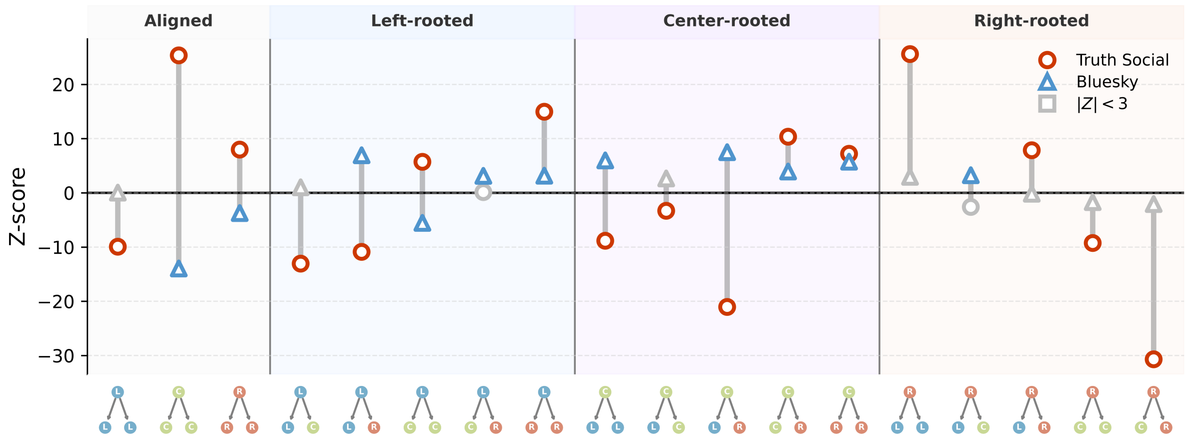


Figure 5: **Star motif analysis.** Standardized Z-scores of three-node star motif frequencies relative to degree-preserving null models. Circles mark Truth Social, triangles Bluesky; gray symbols indicate values with $|Z| < 3$.

design. The two platforms differ substantially in user base size, ideological composition, and patterns of interaction, and they may also differ in unobserved visibility and moderation processes. Our analysis therefore treats platform differences as emergent patterns in observed public interaction traces, rather than as causal effects of technical architecture.

These findings matter because they show that diffusion is not only about reach, speed, or volume, but also about the structure of participation. A cascade can become large through broad, shallow attention or through deeper sequential exchange, and these forms imply different kinds of political engagement. This distinction is especially important in a fragmented platform ecosystem, where newer platforms may share familiar Twitter-like affordances while supporting different conversational dynamics. By comparing reposts and replies separately, our study shows that amplification can scale similarly across platforms even when conversation diverges. Political diffusion should therefore be understood not only as whether information spreads, but as how participation is organized once it does.

Two implications follow. The first concerns what the two cascade shapes mean for how a conversation is organized. Wide, shallow reply cascades consist of many users responding to the same post with little subsequent exchange among them; narrower, deeper cascades involve replies that themselves receive replies across multiple turns. Neither is inherently more desirable, but they distribute attention differently, and the difference arises here between platforms with similar reply mechanics.

The second is methodological. Comparisons based on aggregate activity or cascade size alone would record Truth Social as simply more active and would miss this divergence entirely, because it appears only after conditioning on size. Structural measures of the kind used here are available wherever thread structure is recorded, and separating the diffusion patterns that recur across platforms from those that track the composition of a particular community is what makes findings portable to the next platform.

More broadly, our findings distinguish between diffusion patterns that appear similar across platforms and those that vary with the communities in which they occur. Repost cascades scaled similarly across Truth Social and Bluesky, whereas reply cascades did not. The divergence in reply structure was closely associated with ideological composition and interaction alignment, suggesting that cross-platform differences in political diffusion are most apparent in the organization of conversation rather than in amplification alone.

### 5.1 Limitations

The alignment ratio is computed from the same reply edges that determine cascade depth and breadth. Cross-ideological exchanges may contribute to longer sequential discussions, but sustained conversations may also create conditions under which additional cross-ideological exchanges occur. We therefore interpret ideological alignment as associated with cascade structure rather than as a causal determinant of it.

Three further constraints bear on interpretation. Our measure of ideological alignment is coarse, relying on a simplified typology that may not capture subtler dimensions of identity. While we account for cascade size, content, and user prominence, unobserved platform dynamics may shape interaction in ways we cannot fully disentangle. And we use "technical similarity" in a deliberately narrow sense: both platforms provide Twitter-like affordances for posting, reposting, and threaded replying, which allows us to construct comparable cascade objects, though they differ in feed ranking, moderation, and recommendation architecture (see Appendix N).

These considerations limit the scope of generalization. Our sampling frame is candidate-centered and restricted to one month of the 2024 U.S. presidential campaign. The ideological composition observed here therefore reflects both the platform context and the particular political conversations included in the sample, and should not be interpreted as representative of either platform's full user base. We also cannot determine whether the same structural patterns extend to other political topics or to non-political discussion. Our results therefore support a narrower conclusion: platforms with similar reply mechanics can exhibit different conversational structures conditional on cascade size, and these differences are associated with the ideological composition and interaction patterns observed within those conversations. Assessing how broadly these patterns generalize will require comparable analyses across platforms, countries, time periods, and topics.

## References


Adamic, L. A.; and Glance, N. 2005. The Political Blogosphere and the 2004 U.S. Election: Divided They Blog. In *Proceedings of the 3rd International Workshop on Link Discovery*, LinkKDD '05, 36–43. New York, NY, USA: Association for Computing Machinery.

Alon, U. 2007. Network motifs: theory and experimental approaches. *Nature Reviews Genetics*, 8(6): 450–461.

Bail, C. A.; Argyle, L. P.; Brown, T. W.; Bumpus, J. P.; Chen, H.; Hunzaker, M. B. F.; Lee, J.; Mann, M.; Merhout, F.; and Volfovsky, A. 2018. Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37): 9216–9221.

Barberá, P.; Jost, J. T.; Nagler, J.; Tucker, J. A.; and Bonneau, R. 2015. Tweeting From Left to Right: Is Online Political Communication More Than an Echo Chamber? *Psychological Science*, 26(10): 1531–1542.

Boulianne, S.; Hoffmann, C. P.; and Bossetta, M. 2025. Social media platforms for politics: A comparison of Facebook, Instagram, Twitter, YouTube, Reddit, Snapchat, and WhatsApp. *New Media & Society*, 27(11): 6006–6037.

Boulianne, S.; and Larsson, A. O. 2023. Engagement with candidate posts on Twitter, Instagram, and Facebook during the 2019 election. *New Media & Society*, 25(1): 119–140.

boyd, d.; Golder, S.; and Lotan, G. 2010. Tweet, Tweet, Retweet: Conversational Aspects of Retweeting on Twitter. In *Proceedings of the 43rd Hawaii International Conference on System Sciences*, HICSS '10, 1–10. IEEE.

Chang, H.-C. H.; Druckman, J. N.; Ferrara, E.; and Willer, R. 2025. Liberals and Conservatives Share Information Differently on Social Media. *PNAS Nexus*, 4(7): pgaf206.

Cheng, J.; Adamic, L.; Dow, P. A.; Kleinberg, J. M.; and Leskovec, J. 2014. Can cascades be predicted? In *Proceedings of the 23rd International Conference on World Wide Web*, 925–936. New York, NY: Association for Computing Machinery.

Cinelli, M.; De Francisci Morales, G.; Galeazzi, A.; Quattrociocchi, W.; and Starnini, M. 2021. The echo chamber effect on social media. *Proceedings of the National Academy of Sciences*, 118(9): e2023301118.

Conover, M.; Ratkiewicz, J.; Francisco, M.; Goncalves, B.; Menczer, F.; and Flammini, A. 2011. Political Polarization on Twitter. In *Proceedings of the International AAAI Conference on Web and Social Media*, 89–96. Washington, DC: AAAI Press.

Del Vicario, M.; Vivaldo, G.; Bessi, A.; Zollo, F.; Scala, A.; Caldarelli, G.; and Quattrociocchi, W. 2016. Echo Chambers: Emotional Contagion and Group Polarization on Facebook. *Scientific Reports*, 6(1): 37825.

DeVerna, M. R.; Pierri, F.; Aiyappa, R.; Pacheco, D.; Bryden, J.; and Menczer, F. 2024. How the cascade inference problem distorts information diffusion. arXiv:2410.21554.

Di Martino, E.; Galeazzi, A.; Starnini, M.; Quattrociocchi, W.; and Cinelli, M. 2025. Ideological fragmentation of the social media ecosystem: From echo chambers to echo platforms. *PNAS Nexus*, 4(9): pgaf262.

Easley, D.; and Kleinberg, J. 2010. *Networks, crowds, and markets: reasoning about a highly connected world*. Cambridge, UK: Cambridge University Press.

Egami, N.; Hinck, M.; Stewart, B. M.; and Wei, H. 2023. Using Imperfect Surrogates for Downstream Inference: Design-based Supervised Learning for Social Science Applications of Large Language Models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Failla, A.; and Rossetti, G. 2024. "I'm in the Bluesky Tonight": Insights from a year worth of social data. *PLOS ONE*, 19(11): e0310330.

Flaxman, S.; Goel, S.; and Rao, J. M. 2016. Filter Bubbles, Echo Chambers, and Online News Consumption. *Public Opinion Quarterly*, 80(S1): 298–320.

FORCE11. 2020. The FAIR Data principles. https://force11.org/info/the-fair-data-principles/.

Garimella, K.; De Francisci Morales, G.; Gionis, A.; and Mathioudakis, M. 2018. Political Discourse on Social Media: Echo Chambers, Gatekeepers, and the Price of Bipartisanship. In *Proceedings of the 2018 World Wide Web Conference*, 913–922. Lyon, France: Association for Computing Machinery.

Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J. W.; Wallach, H.; Iii, H. D.; and Crawford, K. 2021. Datasheets for datasets. *Communications of the ACM*, 64(12): 86–92.

Gerard, P.; Botzer, N.; and Weninger, T. 2023. Truth Social Dataset. In *Proceedings of the International AAAI Conference on Web and Social Media*, 1034–1040. Washington, DC: AAAI Press.

Goel, S.; Anderson, A.; Hofman, J.; and Watts, D. J. 2016. The Structural Virality of Online Diffusion. *Management Science*, 62(1): 180–196.

González-Bailón, S.; Lazer, D.; Barberá, P.; Godel, W.; Allcott, H.; Brown, T.; Crespo-Tenorio, A.; Freelon, D.; Gentzkow, M.; Guess, A. M.; Iyengar, S.; Kim, Y. M.; Malhotra, N.; Moehler, D.; Nyhan, B.; Pan, J.; Velasco Rivera, C.; Settle, J.; Thorson, E.; Tromble, R.; Wilkins, A.; Wojcieszak, M.; Kiewiet de Jonge, C.; Franco, A.; Mason, W.; Stroud, N. J.; and Tucker, J. A. 2024. The Diffusion and Reach of (Mis)Information on Facebook during the U.S. 2020 Election. *Sociological Science*, 11: 1124–1146.

Gruhl, D.; Guha, R.; Liben-Nowell, D.; and Tomkins, A. 2004. Information diffusion through blogspace. In *Proceedings of the 13th International Conference on World Wide Web*, 491–501. New York, NY: Association for Computing Machinery.

Guerrero-Solé, F. 2018. Interactive Behavior in Political Discussions on Twitter: Politicians, Media, and Citizens' Patterns of Interaction in the 2015 and 2016 Electoral Campaigns in Spain. *Social Media + Society*, 4(4): 2056305118808776.

Guess, A. M.; Malhotra, N.; Pan, J.; Barberá, P.; Allcott, H.; Brown, T.; Crespo-Tenorio, A.; Dimmery, D.; Freelon, D.; Gentzkow, M.; González-Bailón, S.; Kennedy, E.; Kim, Y. M.; Lazer, D.; Moehler, D.; Nyhan, B.; Rivera, C. V.; Settle, J.; Thomas, D. R.; Thorson, E.; Tromble, R.; Wilkins, A.; Wojcieszak, M.; Xiong, B.; and de Jonge, C. K. 2023a. Reshares on social media amplify political news but do not detectably affect beliefs or opinions. *Science*, 381(6656): 404–408.

Guess, A. M.; Malhotra, N.; Pan, J.; Barberá, P.; Allcott, H.; Brown, T.; Crespo-Tenorio, A.; Dimmery, D.; Freelon, D.; Gentzkow, M.; González-Bailón, S.; Kennedy, E.; Kim, Y. M.; Lazer, D.; Moehler, D.; Nyhan, B.; Rivera, C. V.; Settle, J.; Thomas, D. R.; Thorson, E.; Tromble, R.; Wilkins, A.; Wojcieszak, M.; Xiong, B.; de Jonge, C. K.; Franco, A.; Mason, W.; Stroud, N. J.; and Tucker, J. A. 2023b. How do social media feed algorithms affect attitudes and behavior in an election campaign? *Science*, 381(6656): 398–404.

Guille, A.; Hacid, H.; Favre, C.; and Zighed, D. A. 2013. Information diffusion in online social networks: a survey. *SIGMOD Rec.*, 42(2): 17–28.

Hughes, E. M.; and Weninger, T. 2025. Echo Chambers and Information Brokers on Truth Social: A Study of Network Dynamics and Political Discourse. In *Companion Publication of the 2025 Conference on Computer-Supported Cooperative Work and Social Computing*, CSCW Companion '25, 292–296. New York, NY, USA: Association for Computing Machinery.

Juul, J. L.; and Ugander, J. 2021. Comparing information diffusion mechanisms by matching on cascade size. *Proceedings of the National Academy of Sciences*, 118(46): e2100786118.

Kim, J.; and Yoo, J. 2012. Role of Sentiment in Message Propagation: Reply vs. Retweet Behavior in Political Communication. In *Proceedings of the 2012 ASE International Conference on Social Informatics, SocialInformatics 2012*, 131–136. Washington, D.C., United States: IEEE Computer Society.

Mosleh, M.; Allen, J.; and Rand, D. G. 2025. Divergent patterns of engagement with partisan and low-quality news across seven social media platforms. *Proceedings of the National Academy of Sciences*, 122(44): e2425739122.

Niitsuma, T.; Yoshida, M.; Tamori, H.; and Nakawake, Y. 2025. Prestige bias drives the viral spread of content reposted by influencers in online communities. arXiv:2411.05448.

Nogara, G.; Sahneh, E. S.; DeVerna, M. R.; Liu, N.; Luceri, L.; Menczer, F.; Pierri, F.; and Giordano, S. 2026. A longitudinal analysis of misinformation, polarization and toxicity on Bluesky after its public launch. *Online Social Networks and Media*, 51: 100342.

Park, C. S.; and Kaye, B. K. 2019. Expanding Visibility on Twitter: Author and Message Characteristics and Retweeting. *Social Media + Society*, 5(2).

Quelle, D.; and Bovet, A. 2025. Bluesky: Network topology, polarization, and algorithmic curation. *PLOS ONE*, 20(2): e0318034.

Rathje, S.; Van Bavel, J. J.; and van der Linden, S. 2021. Out-group animosity drives engagement on social media. *Proceedings of the National Academy of Sciences*, 118(26): e2024292118.

Rowe, M.; and Alani, H. 2014. Mining and comparing engagement dynamics across multiple social media platforms. In *Proceedings of the 2014 ACM conference on Web science*, 229–238. New York, NY: Association for Computing Machinery.

Shah, K.; Gerard, P.; Luceri, L.; and Ferrara, E. 2024. Unfiltered Conversations: A Dataset of 2024 U.S. Presidential Election Discourse on Truth Social. HUMANS Lab Working Paper No. 2024.8, arXiv:2411.01330.

Shugars, S.; and Beauchamp, N. 2019. Why Keep Arguing? Predicting Engagement in Political Conversations Online. *SAGE Open*, 9(1).

Slaughter, I.; Peytavin, A.; Ugander, J.; and Saveski, M. 2025. Community notes reduce engagement with and diffusion of false information online. *Proceedings of the National Academy of Sciences*, 122(38): e2503413122.

TeBlunthuis, N.; Hase, V.; and Chan, C.-H. 2024. Misclassification in Automated Content Analysis Causes Bias in Regression. Can We Fix It? Yes We Can! *Communication Methods and Measures*, 18(3): 278–299.

Toraman, C.; Şahinuç, F.; Yilmaz, E. H.; and Akkaya, I. B. 2022. Understanding Social Engagements: A Comparative Analysis of User and Text Features in Twitter. *Social Network Analysis and Mining*, 12(1): 47.

Vallejo Vera, S.; and Driggers, H. 2025. LLMs as annotators: the effect of party cues on labelling decisions by large language models. *Humanities and Social Sciences Communications*, 12: 1530.

Vosoughi, S.; Roy, D.; and Aral, S. 2018. The spread of true and false news online. *Science*, 359(6380): 1146–1151.

Wang, X.; Chao, F.; Yu, G.; and Zhang, K. 2022. Factors influencing fake news rebuttal acceptance during the COVID-19 pandemic and the moderating effect of cognitive ability. *Computers in Human Behavior*, 130: 107174.

Zannettou, S.; Bradlyn, B.; De Cristofaro, E.; Kwak, H.; Sirivianos, M.; Stringhini, G.; and Blackburn, J. 2018. What is Gab? A Bastion of Free Speech or an Alt-Right Echo Chamber? In *Companion Proceedings of the Web Conference 2018*, WWW '18 Companion, 1007–1014. New York, NY, USA: Association for Computing Machinery.

Zhao, Z.; Zhao, J.; Sano, Y.; Levy, O.; Takayasu, H.; Takayasu, M.; Li, D.; Wu, J.; and Havlin, S. 2020. Fake news propagates differently from real news even at early stages of spreading. *EPJ Data Science*, 9(1): 1–14.

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   Yes.
   (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope?
   Yes.
   (c) Do you clarify how the proposed methodological approach is appropriate for the claims made?
   Yes.
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   Yes.
   (e) Did you describe the limitations of your work?
   Yes.
   (f) Did you discuss any potential negative societal impacts of your work?
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   Yes.
   (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings?
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   (c) Did you include any new assets in the supplemental material or as a URL?
   Yes, subject to platform terms and privacy constraints.
   (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating?
   Yes. The study uses publicly available social media data and does not involve direct interaction with users.
   (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content?
   Yes.

(f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see FORCE11 (2020))?
Not applicable.

(g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al. (2021))?
Not applicable.

6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...

(a) Did you include the full text of instructions given to participants and screenshots?
Not applicable.

(b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals?
Not applicable.

(c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation?
Not applicable.

(d) Did you discuss how data is stored, shared, and deidentified?
Yes.

# A Power-Law Analysis of User-Level Activity Distributions

We analyzed the distribution of user-level activity—specifically, the number of followers, reposts, and replies per user—across both platforms using power-law fits. Figure 6 shows the complementary cumulative distribution functions (CCDFs) of these metrics in log-log scale, overlaid with fitted power-law tails (dashed orange lines). For each distribution, we report the estimated power-law exponent $\gamma$ and the lower cutoff $x_{\min}$ above which the power-law holds.

The distribution of follower counts reveals strong skewness on both platforms. On Truth Social, a small number of users amass an exceptionally large following, with $\gamma = 1.76$ and $x_{\min} \approx 13{,}515$. Bluesky shows a slightly less extreme concentration ($\gamma = 1.84$, $x_{\min} \approx 1{,}498$). Repost and reply behaviors also exhibit heavy-tailed patterns. On Bluesky, replies decay rapidly ($\gamma = 3.02$) while reposts follow a shallower slope ($\gamma = 2.43$). Truth Social displays flatter distributions for both reposts ($\gamma = 1.84$) and replies ($\gamma = 1.79$), consistent with broader participation from users in redistributing and responding to content.

### Exclusion of Highly Followed Users for Robustness.

To identify unusually influential users, we examined the follower-count distribution of every account in our data (134,100 on Truth Social; 41,566 on Bluesky). Both distributions are heavy tailed, but their upper tails differ in kind. On Truth Social the top eight accounts are separated from the body of the distribution: the largest (Donald Trump, 7,107,831 followers) has 56 times the followers of the platform's 99.9th-percentile account (127,082), the eighth-largest still holds 2,170,535 followers, and together these eight hold 16.4% of the total follower count. On Bluesky no comparable class exists: the largest account (1,230,128) sits five times above the 99.9th percentile (238,048), continuous with the rest of the distribution, and the top eight hold 6.1% of the total. The eighth-largest Truth Social account exceeds the largest Bluesky account.

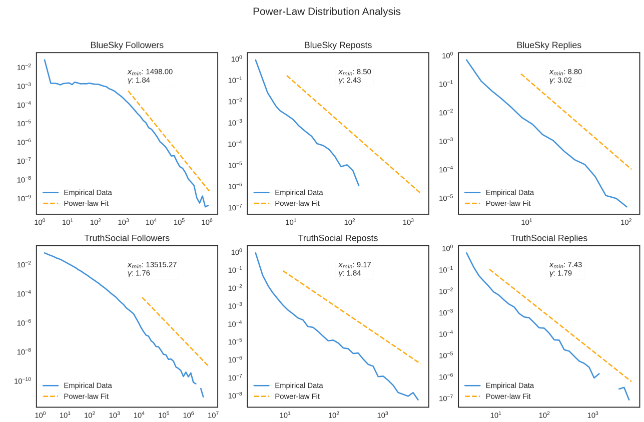


Figure 6: Follower count distributions across platforms.

We therefore designate the eight separated Truth Social accounts as influencers and apply no exclusion on Bluesky. A symmetric rule, such as excluding the top $n$ percent on both platforms, would not equalize the comparison: on Bluesky it would remove accounts continuous with the rest of the distribution, while on Truth Social any $n$ large enough to capture the separated class would also sweep in ordinary accounts.

# B Ideology Estimation: Validation and Robustness

### Ideological Distribution of Root Post Authors.

Figure 7 shows the ideological distribution of users who authored root posts. Bluesky exhibited a predominantly left-leaning user base (over 46,000 root posts from left-leaning users vs. 27,495 from right-leaning), while Truth Social was overwhelmingly right-leaning (37,639 right-authored root posts vs. 5,149 from left-leaning users and 1,053 from centrists).

### Robustness to Ideology Thresholds.

We re-estimated Model 3 using alternative thresholds of 0.5 and 0.7 for classifying user ideology. As shown in Table 2 and Figure 8, the results remain substantively consistent across thresholds: pseudo-$R^2$ values are uniformly high, error metrics fluctuate only marginally, and the platform–size interaction is small at every threshold once ideological structure is incorporated. The residual interaction shrinks as the threshold rises, from 0.155 and $-0.107$ at 0.5 to $-0.008$ and 0.013 at 0.7, because a stricter threshold sorts more users into the center category and composition absorbs more of

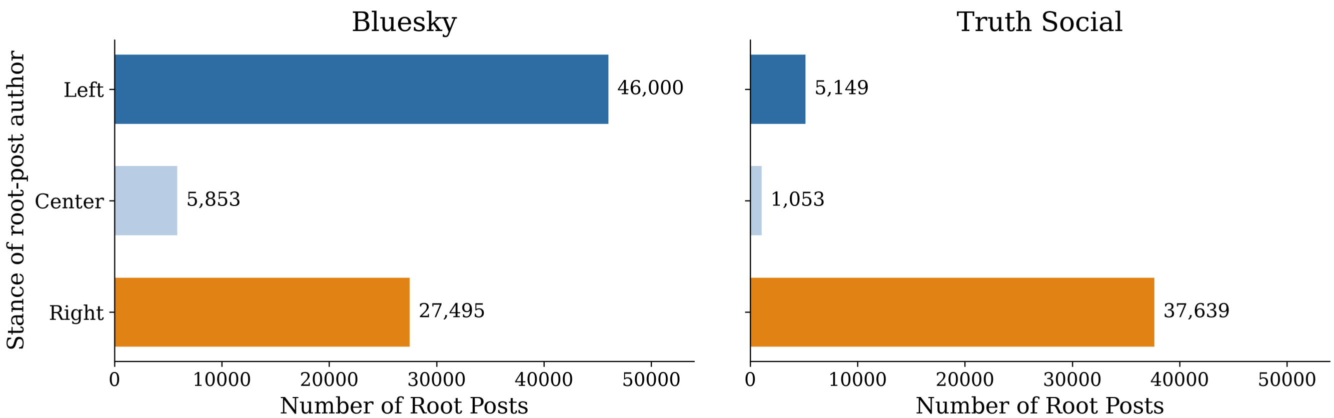


Figure 7: Distribution of root post partisanship across platforms. Bar plots show the number of root posts authored by users with left-leaning, centrist, or right-leaning stance labels. On *Bluesky*, left-leaning users dominate root-post activity. *Truth Social* exhibits a pronounced right-leaning skew. These distributions reflect the ideological clustering of user bases on each platform.

the platform contrast. The conclusion holds throughout. We also applied an unsupervised natural breakpoint detection algorithm to the ideology distribution, which identified 0.67 as a statistically distinct split point; we adopt 0.6 as our default to avoid excessive sparsity.

### Class-Specific Validation Uncertainty.

Inter-annotator agreement on the 200-item validation sample is substantial (Cohen's $\kappa = 0.77$, bootstrap 95% interval $[0.70, 0.85]$), and model–human agreement is moderate ($\kappa = 0.61$ with Annotator 1 and 0.74 with Annotator 2). Table 3 reports class-specific and platform-specific precision and recall against the 171 items on which both annotators agree, with bootstrap intervals. Overall accuracy against consensus is 0.86.

The errors are not uniform across classes, and the pattern matters for interpreting our composition estimates: the model over-assigns each platform's ideological minority. Precision is 0.64 for right-leaning labels on Bluesky and 0.64 for left-leaning labels on Truth Social, against 0.98 and 0.93 for the largest class on each platform. Because the minority-class estimates rest on relatively few consensus cases, we distinguish between the direction of the observed bias and its precise magnitude: the two minority cells rest on 8 and 17 consensus items and their intervals are correspondingly wide (for right on Bluesky, $[0.33, 0.91]$). Bootstrapping the difference between minority-class and majority-class precision within each platform gives $-0.34$ on Bluesky (one-sided $p = 0.017$) and $-0.29$ on Truth Social ($p = 0.004$), so the direction is supported on both platforms. Recomputing against each annotator alone gives lower precision throughout but leaves the ordering unchanged. Uneven performance across the ideological spectrum is a documented property of language-model annotators (Vallejo Vera and Driggers 2025).

### Robustness to Ideology Label Noise.

Because the stance labels contain measurement error, we tested whether the central result survives label noise at the measured rates. Misclassification by automated classifiers biases downstream regression even when overall accuracy is high (TeBlunthuis, Hase, and Chan 2024), and the same holds for language-model labels used as surrogates (Egami et al. 2023). Those authors correct the bias analytically with a gold-standard subsample; we instead propagate the measured error through the full pipeline, because our labels enter through composition, alignment, splines, and interactions rather than as a single regressor. For each of 100 draws per scenario, every user's label was replaced by a draw from the measured confusion distribution, cascade-level composition and alignment were recomputed, and Model 3c was refit.

We ran this under four error processes: the measured platform-specific rates; a pooled matrix that applies each platform's dominant class the error rate measured largely on the other platform; a nested bootstrap that resamples the validation items and rebuilds the confusion matrix on each draw, carrying the uncertainty in Table 3 through; and the harsher rates measured against a single annotator, under which precision for right-leaning labels on Bluesky falls to 0.33. Relative to the baseline interactions without ideology controls ($0.398$ and $-0.334$), the median draw absorbs between 86% and 96% of the platform divergence, and the least favorable of the four hundred draws absorbs 70%. No error process we simulated restores the platform gap. All fits here share one scale estimate and compare perturbed against unperturbed values, so the intervals are ranges across simulation draws rather than standard errors.

### The Center Category.

The center label is a residual, mixing genuinely moderate users, users who post about politics inconsistently, and users whose stance the classifier could not resolve. To test whether this residual carries the explanatory work, we recomputed composition and alignment using partisan users only, so that center-labeled users enter as unlabeled; alignment is then

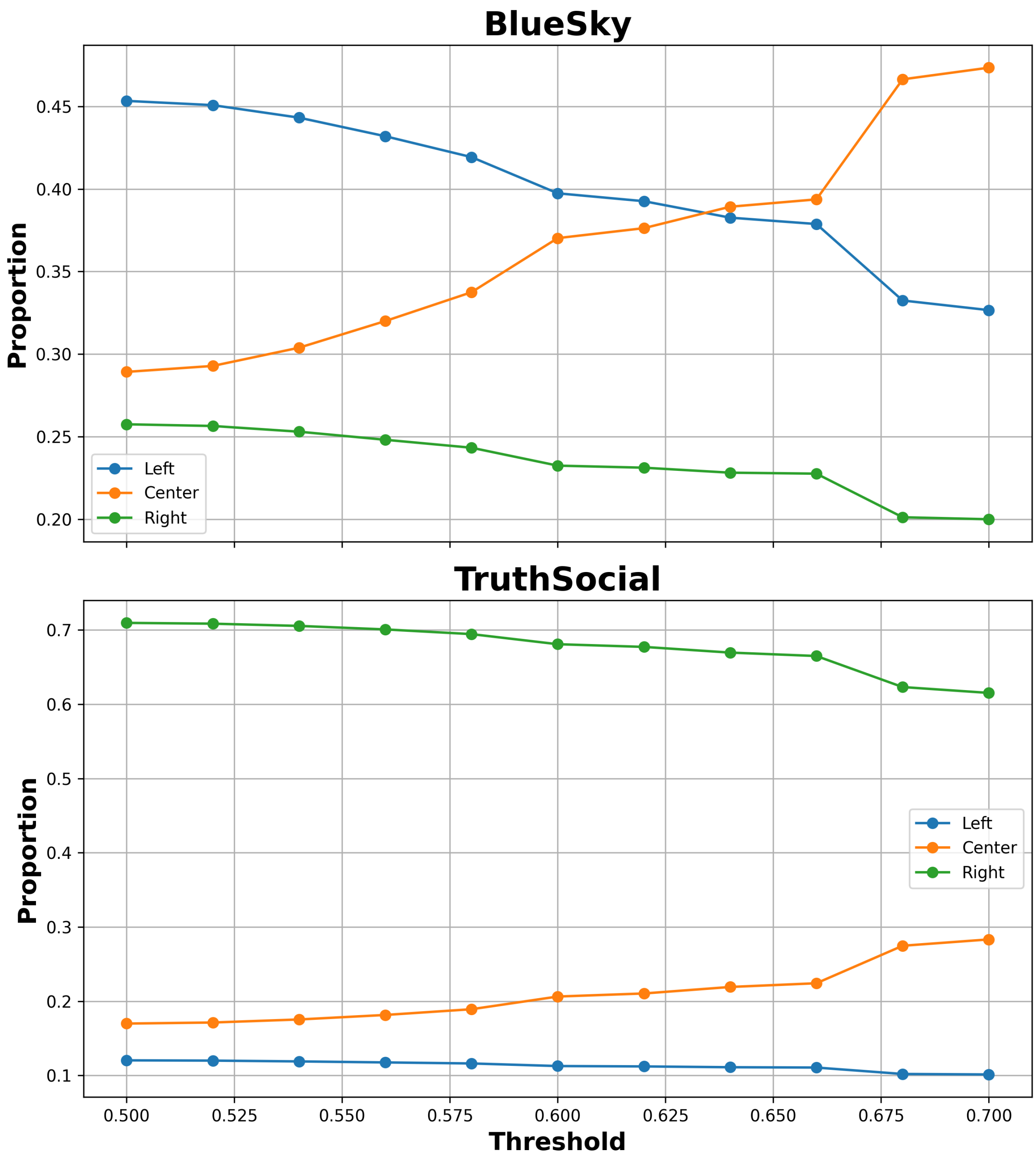


Figure 8: User ideology proportions and consistency across thresholds on *Bluesky* and *Truth Social*. Left-leaning users dominate at lower thresholds on Bluesky; right-leaning users represent the majority on Truth Social at all thresholds. Center-labeled users increase at higher thresholds on both platforms.

Table 2: Robustness check using alternative thresholds for defining user-level ideology. Results show that the main findings are stable across thresholds (0.5, 0.6, 0.7): platform gaps in cascade breadth and depth are consistently reduced once ideological structure is accounted for. Entries are Model 3c (composition and alignment jointly) fitted with Huber's proposal 2 scale, matching Table 1; $n = 120{,}693$ cascades for which reply-tree ideology shares could be recomputed at each threshold.

| | Breadth (Log) | | | | Depth (Log) | | | |
|---|---|---|---|---|---|---|---|---|
| Threshold | Pseudo-$R^2$ | MAE | RMSE | Size×TS | Pseudo-$R^2$ | MAE | RMSE | Size×TS |
| 0.5 | 0.954 | 0.044 | 0.100 | 0.155 | 0.908 | 0.039 | 0.084 | -0.107 |
| 0.6 | 0.955 | 0.047 | 0.098 | 0.066 | 0.908 | 0.040 | 0.084 | -0.045 |
| 0.7 | 0.957 | 0.045 | 0.096 | -0.008 | 0.913 | 0.038 | 0.082 | 0.013 |

| Scope | Class | Precision [95% CI] | Recall | $n$ |
|---|---|---|---|---|
| Pooled | left | 0.68 [0.54, 0.82] | 0.80 | 35 |
| Pooled | center | 0.96 [0.90, 1.00] | 0.85 | 59 |
| Pooled | right | 0.89 [0.81, 0.95] | 0.90 | 77 |
| Bluesky | left | 0.74 [0.53, 0.93] | 0.78 | 18 |
| Bluesky | center | 0.98 [0.92, 1.00] | 0.89 | 45 |
| Bluesky | right | 0.64 [0.33, 0.91] | 0.88 | 8 |
| Truth Social | left | 0.64 [0.42, 0.83] | 0.82 | 17 |
| Truth Social | center | 0.91 [0.69, 1.00] | 0.71 | 14 |
| Truth Social | right | 0.93 [0.86, 0.98] | 0.90 | 69 |

Table 3: Class-specific and platform-specific model performance against human consensus labels (171 of 200 validation items on which both annotators agree). Intervals are percentile bootstrap 95% intervals over 2,000 resamples; $n$ is the consensus support per class.

measured over reply edges whose two endpoints are both partisan. On the resulting subset of 28,083 cascades, composition and alignment still absorb 61% of the baseline breadth divergence and 64% of the depth divergence, compared with 99% and 77% under the published specification on the same sample. Under the partisan-only specification, ideological composition and alignment still account for a substantial, although smaller, share of the platform divergence. Part of the accounting in the three-category specification depends on separating users the classifier places confidently on one side from the residual center category; because center collects moderate, inconsistent, and unresolved users alike, this contrast is not the same as one between partisan and non-partisan participation.

### Ideological Composition Compared With Prior Estimates.

Our labels place 41.4% of Bluesky users in our sample on the left, 34.4% at center, and 24.2% on the right; on Truth Social the shares are 11.5% left, 19.3% center, and 69.2% right. The Bluesky figures show a larger right-leaning presence than platform-wide estimates: Quelle and Bovet (2025) classify 75.3% of Bluesky users as left of center and 4.8% as right of center, and Nogara et al. (2026), scoring active users by the leaning of the domains they share, report 74.61% left-leaning, 18.17% centrist, and 7.21% right-leaning.

Two differences account for most of this gap. First, the samples differ: platform-wide estimates describe all users, most of whom post little political content, and are typically derived from shared news domains, whereas our sample is restricted to accounts posting in a conversation mentioning Biden or Trump during one campaign month, labeled from post text. Second, part of the gap is measurement error. Applying a prevalence correction that inverts the measured confusion matrix lowers the estimated right-leaning share on Bluesky from 24.2% to 17.9% and the left-leaning share on Truth Social from 11.5% to 1.9%. Both corrections move in the expected direction, but for Bluesky a substantial gap remains after correction, so label error alone does not account for it. The remaining discrepancy may partly reflect differences in sampling frame, since a candidate-centered conversation sample is not a sample of the platform population. These corrected values are indicative rather than precise, because the correction inherits the imprecision of the minority cells in Table 3.

## C Descriptive Statistics

Table 4 reports cascade counts, post counts, and the cascade size distribution for both platforms and both cascade types.

## D Robustness of Repost Cascade Reconstruction

Any reconstruction rule could in principle shape the structural comparison (DeVerna et al. 2024), so we re-derived every repost cascade under alternative rules and re-estimated the platform-by-size interaction ($\beta_3$ in Eq. 1) for each. Every admissible rule selects each repost's parent from the same candidate set: the original post, plus earlier reposters whom the reposting user follows. We compare the rule used in the paper against the earliest eligible reposter, the most recent eligible reposter, and a uniformly random eligible reposter (40 draws), each under both readings of the repost list order.

Figure 9 summarizes the result over 244,129 repost cascades (4.03 million reposts). No rule yields a breadth interaction outside $[-0.012, +0.031]$ or a depth interaction outside $[-0.083, +0.028]$. The same estimator applied to reply cascades gives $+0.170$ (breadth) and $-0.182$ (depth), exceeding the most extreme repost estimate under any rule by a factor of five and two respectively. The original reconstruction rule yields the largest estimated platform difference among the specifications considered; every alternative moves the repost estimates closer to zero. The similarity in repost scaling is therefore robust to the reconstruction rules examined here. (Because log breadth and log depth of small

| Type | Platform | Cascades | Posts | Share of cascades by size (%) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 1 | 2–5 | 6–50 | >50 |
| Reply | Bluesky | 79,348 | 199,413 | 71.3 | 21.2 | 7.2 | 0.3 |
| Reply | Truth Soc. | 43,841 | 1,369,497 | 43.3 | 25.2 | 22.5 | 9.0 |
| Repost | Bluesky | 25,550 | 184,807 | 46.3 | 33.6 | 17.8 | 2.3 |
| Repost | Truth Soc. | 218,579 | 3,847,903 | 51.4 | 29.7 | 13.9 | 5.0 |

Table 4: Cascade counts and size distributions. For reply cascades, size is the number of posts in the thread including the root. For repost cascades, size is the number of reposts.

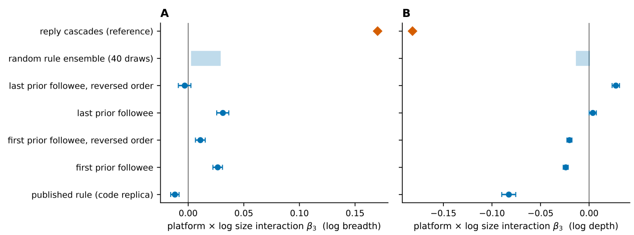


Figure 9: Platform-by-size interaction $\beta_3$ for repost cascades under alternative reconstruction rules (dots with 95% confidence intervals; shaded band, range over 40 random-rule draws), for log breadth (A) and log depth (B). The diamond marks the reply-cascade interaction under the same estimator.

trees take few distinct values, the scale estimate of a Huber robust regression degenerates on repost cascades; the interactions in this analysis are estimated by OLS with HC3 standard errors.)

The rule has limited room to matter in the first place: 55% of Bluesky reposts and 59% of Truth Social reposts have no eligible prior reposter and attach to the root under every rule, and a further 21% and 12% have exactly one candidate. A second concern is that the follower network is a post-collection snapshot without edge creation dates, so a follow created after a repost could produce a spurious parent. Deleting follow edges at random at rates from 5% to 30% to simulate such edges moves the breadth interaction only from $0.031$ to $0.005$ and the depth interaction from $0.004$ to $0.013$, both far below the reply values. Both follow networks were collected at the same time, so this source of error applies comparably to the two platforms.

## E Unit of Analysis: Posts Versus Users

Reply cascades are trees of posts, so a user who replies several times appears once per reply. An alternative representation collapses each user to a single node at the shallowest depth at which they appear. We recomputed all reply-cascade metrics both ways from the raw thread data. The post-level reconstruction reproduces the metrics used throughout the paper (correlations of $0.999$ for size, $0.999$ for breadth, $0.988$ for depth), confirming the reported analyses are post-level.

The two representations are compared to each other within a common subsample, not to Table 1. Both estimates are restricted to cascades that actually contain a reply, since a lone root post has the same structure under either representation: 47,106 cascades at post level, and 42,947 once repeat posters are merged and single-participant threads drop out. Coefficients are therefore not comparable in level to the main table, whose baseline $\beta_3$ of $0.419$ and $-0.321$ is estimated on all 123,189 cascades, 61% of which are single posts. Within this subsample, estimated with the same Huber proposal 2 specification, the post-level baseline is $0.168$ for breadth and $-0.194$ for depth.

Against those subsample baselines, the depth divergence persists under user collapsing ($-0.132$, against $-0.194$ at post level) and is again largely absorbed by composition and alignment. The breadth divergence is much smaller under collapsing ($-0.032$, against $0.168$). A substantial part of the raw breadth gap therefore reflects the same Truth Social users posting repeatedly within a thread, which widens a level of the post tree without adding distinct participants. Conditional on size, Truth Social conversations are wider in posts, and part of that width comes from repeat participation rather than from a broader set of participants. The depth contrast holds under both representations.

## F Two-User Exchanges in Reply Cascades

Depth would mean something different if it largely measured back-and-forth exchange between the same two accounts rather than extended multi-party discussion. We therefore counted, for every reply, whether its author also wrote its grandparent post, that is, an $A \to B \to A$ pattern. Among all cascades, 9.3% on Bluesky and 20.8% on Truth Social contain at least one such exchange, reflecting mainly that Truth Social cascades are larger. Conditioning on cascades of depth two or more, the share is 68.8% on Bluesky and 70.0% on Truth Social, and such exchanges account for 20.0% and 15.6% of reply edges respectively. Dyadic back-and-forth is therefore common on both platforms to a very similar degree, so the depth difference we report is not explained by one platform hosting more two-user conversations.

## G Motif Frequencies and the Null Model

Table 5 reports the raw observed frequency of every ideology-labeled three-node motif on each platform, alongside which the z-scores should be read. The two platforms differ by orders of magnitude in absolute motif volume, so a large z-score does not imply a common configuration.

| | Star | | Chain | |
|---|---|---|---|---|
| Ideology triple | Bluesky | Truth Soc. | Bluesky | Truth Soc. |
| L–L–L | 13,436 | 1,423,162 | 6,519 | 4,747 |
| L–L–C | 20,482 | 1,361,062 | 2,659 | 3,270 |
| L–L–R | 3,869 | 4,721,187 | 346 | 11,883 |
| L–C–L | 18,970 | 1,695,796 | 4,966 | 3,727 |
| L–C–C | 31,468 | 2,929,973 | 3,113 | 7,194 |
| L–C–R | 6,179 | 6,432,294 | 290 | 8,869 |
| L–R–L | 3,751 | 4,837,988 | 471 | 26,476 |
| L–R–C | 5,846 | 5,384,674 | 191 | 10,673 |
| L–R–R | 1,462 | 20,194,975 | 88 | 40,264 |
| C–L–L | 16,997 | 158,383 | 1,533 | 2,040 |
| C–L–C | 36,051 | 137,961 | 6,197 | 3,390 |
| C–L–R | 8,058 | 437,959 | 249 | 5,929 |
| C–C–L | 30,243 | 142,810 | 2,930 | 3,421 |
| C–C–C | 74,502 | 197,087 | 25,971 | 13,261 |
| C–C–R | 15,559 | 458,039 | 980 | 8,179 |
| C–R–L | 6,786 | 466,118 | 161 | 7,371 |
| C–R–C | 15,173 | 458,308 | 1,518 | 8,093 |
| C–R–R | 4,093 | 1,929,621 | 171 | 16,393 |
| R–L–L | 407 | 11,197,554 | 89 | 15,565 |
| R–L–C | 841 | 6,961,901 | 111 | 14,127 |
| R–L–R | 281 | 24,241,501 | 307 | 66,065 |
| R–C–L | 778 | 8,183,288 | 111 | 15,549 |
| R–C–C | 1,735 | 6,435,618 | 384 | 30,188 |
| R–C–R | 625 | 19,234,403 | 819 | 48,366 |
| R–R–L | 238 | 23,352,704 | 74 | 45,060 |
| R–R–C | 448 | 15,651,387 | 163 | 30,149 |
| R–R–R | 224 | 64,855,781 | 357 | 128,058 |
| Total | 318,502 | 233,481,534 | 60,768 | 578,307 |

Table 5: Observed counts of ideology-labeled three-node motifs (L = left, C = center, R = right). For chains the triple is ordered along the reply path; for stars it lists the root followed by the two leaves.

Motifs of this kind are not mutually independent: a chain of four nodes contains two overlapping three-node chains. This dependence does not bias the z-scores, because the null model is subjected to the identical counting procedure. We generate 100 randomized graphs by double-edge swaps performed within each cascade: two reply edges are chosen at random and their targets exchanged, together with the target's ideology label, with $5m$ swap attempts in a cascade of $m$ edges. This holds fixed the set of cascades and their sizes, each node's out-degree, and the ideology attached to every participant, while randomizing who replies to whom. What the dependence does affect is the joint interpretation of several motifs at once: neighboring motif types share substructure, so their z-scores are correlated and we read them as a pattern across families rather than as independent evidence.

## H Topic Model Details

The topic model was fit over 125,623 root posts pooled across platforms. HDBSCAN initially assigns low-density documents to an outlier class; we applied BERTopic's outlier reduction step with default settings, which reassigns those documents to their nearest topic, leaving 21 documents (0.02%) unassigned. Clustering yields 12 topics plus a residual outlier class. Manual labeling then produced the 11 coherent and interpretable categories used in the analysis: the residual class, which retains only 21 posts, is excluded, and two near-duplicate pro-Trump categories are treated as one. Every cascade in the modeling sample carries one of these 11 labels. Because topic variation accounts for relatively little of the platform difference (Model 4), uncertainty introduced by outlier reassignment is unlikely to affect the main comparison.

## I Modeling Strategy and Regression Diagnostics

Initial Ordinary Least Squares (OLS) regression revealed strong heteroskedasticity, with residuals showing a fan-shaped pattern as a function of fitted values (Figure 10). Residuals also exhibited heavy tails resembling a t-distribution. Although Cook's Distance analysis identified a small number of observations with relatively higher influence, the vast majority fell well below the threshold of $4/n$. The heavy-tailed, heteroskedastic residuals motivate robust estimation, and all models reported in the main text are robust linear models with the Huber loss. Because 61.3% of reply cascades consist of a root post with no replies, and these sit exactly at the origin of the log–log scaling plot, the default median-absolute-deviation scale estimate collapses toward zero; standard errors computed from it are not interpretable. We therefore fit with Huber's proposal 2 scale, which does not degenerate here, and report a sensitivity analysis across samples and scale estimators below. The one analysis estimated by OLS with HC3 errors is the repost reconstruction robustness check, where lattice-valued outcomes make the Huber scale estimate degenerate.

### Sensitivity of the Modeling Specification.

Two features of the reply-cascade data warrant an explicit check. First, 61.3% of reply cascades consist of a root post with no replies (71.3% on Bluesky, 43.3% on Truth Social). These are structurally degenerate: breadth and depth are fixed by definition, so they sit exactly at the origin of the log–log scaling plot. Second, because this mass of exactly-fitted points makes the median absolute deviation of the residuals vanish, the default scale estimator collapses toward zero, and standard errors computed from it are not interpretable.

Table 6 therefore re-estimates the baseline model and Model 3c across the four combinations of sample and scale estimator. The baseline interaction is smaller when single-post cascades are excluded, as expected once the degenerate points are removed, but the substantive result is stable: including composition and alignment absorbs between 81% and 94% of the baseline platform difference in every specification, for both breadth and depth. Table 1 reports the Huber proposal 2 fit on all cascades.

## J Spline Transformations

To examine potential nonlinear predictor effects, we inspected partial residual plots for the three ratio predictors (Figure 11). Each ratio enters the model both on its own

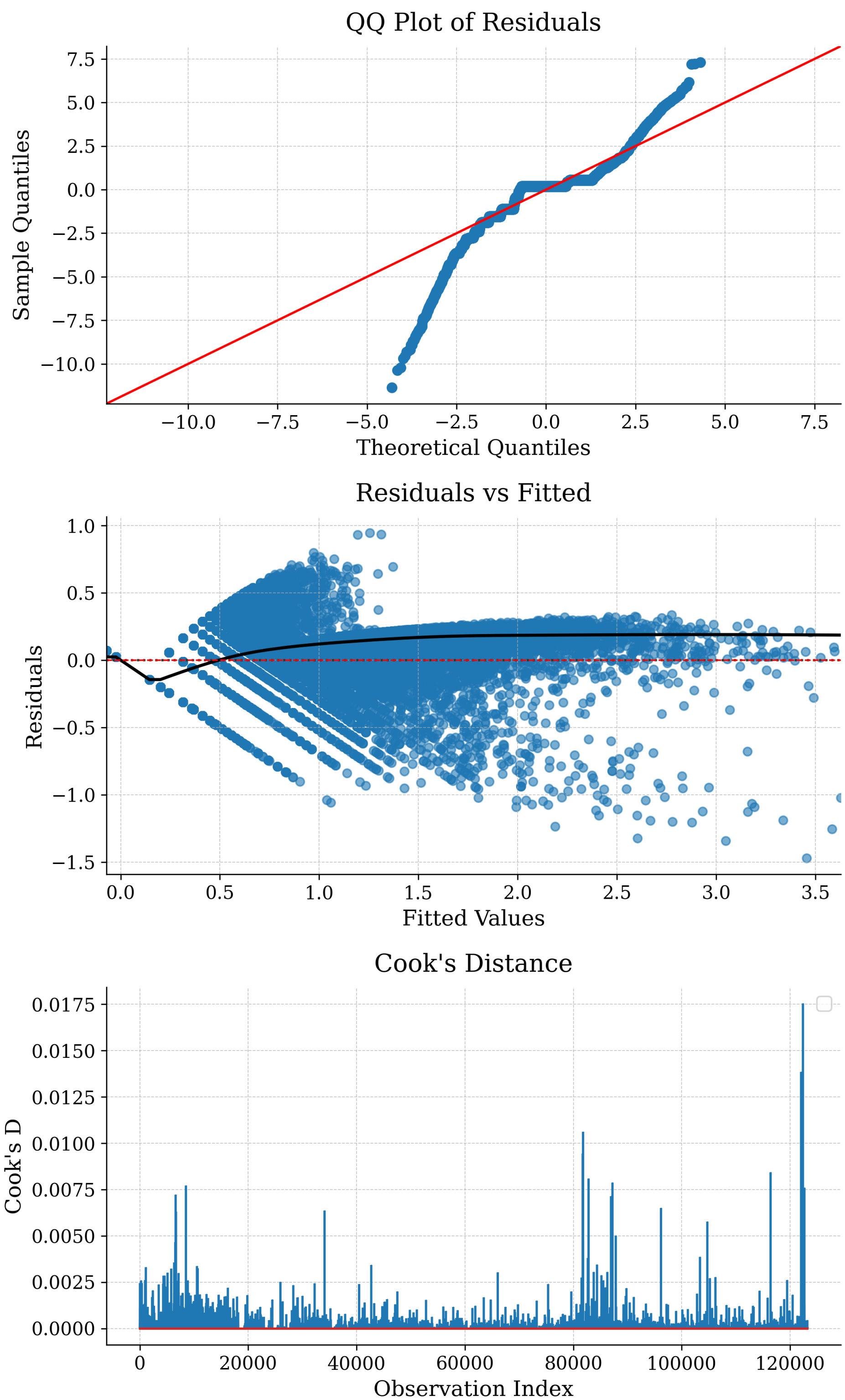


Figure 10: **Regression diagnostic plots.** Top: quantile–quantile (QQ) plot of residuals indicating heavy-tailed error distribution. Bottom: residuals vs. fitted values showing a fan-shaped pattern (heteroskedasticity). LOESS smoother overlaid.

| Sample | Scale est. | Outcome | Baseline | Model 3c | Absorbed |
|---|---|---|---|---|---|
| All | MAD | breadth | 0.398 | 0.023 | 94% |
| All | MAD | depth | $-0.334$ | $-0.055$ | 84% |
| All | Huber | breadth | 0.419 | 0.071 | 83% |
| All | Huber | depth | $-0.321$ | $-0.049$ | 85% |
| Size $> 1$ | MAD | breadth | 0.187 | 0.035 | 81% |
| Size $> 1$ | MAD | depth | $-0.223$ | 0.019 | 91% |
| Size $> 1$ | Huber | breadth | 0.181 | 0.015 | 92% |
| Size $> 1$ | Huber | depth | $-0.217$ | 0.029 | 87% |

Table 6: Platform-by-size interaction before and after adding ideological composition and alignment, across samples and scale estimators. "Absorbed" is the proportion of the baseline interaction removed by Model 3c. $n = 123{,}189$ for all cascades and 47,670 for cascades with more than one post.

and interacted with $\log(\text{size})$, so each panel plots the predictor's full contribution against a straight line carrying the slope that contribution has at mean cascade size. Departure of the smoother from that line is what a spline would capture.

The left-user ratio tracks its linear term closely across the range. The right-user ratio departs from it between roughly 0.6 and 0.95, and the alignment ratio departs from it across most of the range. The largest gap between smoother and line is 0.06 for the left-user ratio, 0.17 for the right-user ratio, and 0.23 for the alignment ratio.

We apply spline transformations (`bs` basis, df = 3) to `author_left_ratio` and `author_alignment_ratio`; B-splines are preferred over natural cubic splines here because the data are confined to [0,1] and used in interactions, where their local support provides greater numerical stability. This places a spline on the predictor with the least curvature and a linear term on one with more, so we report what the choice costs. Refitting Model 3c with all three ratios splined absorbs 98.8% of the baseline breadth interaction and 93.5% of the depth interaction; entering all three linearly absorbs 68.4% and 69.7%. The specification we report falls between these, at 83.1% and 84.7%. It is therefore the more conservative of the two spline choices, and the finding that ideological structure absorbs most of the platform gap holds wherever the splines are placed.

## K Final Model Specification

Our final model specification includes: (1) a main effect of `log_size` and its interaction with `platform`; (2) `right_user_ratio` and spline-based transformations of `alignment_ratio` and `left_user_ratio`, interacted with both `log_size` and `platform`; and (3) topic category and outlier status as categorical moderators. All models were estimated using the `statsmodels` package in Python with robust standard errors.

## L Robustness Checks: Alternative User Composition Specifications

To assess whether findings depend on how user composition is encoded, we compared four specifications: (1) left/right shares only, (2) majority/minority shares only, (3) left/right with alignment, and (4) majority/minority with alignment. Table 7 reports results for both cascade breadth and depth.

Including alignment substantially improves fit for both outcomes (Breadth $R^2 \approx 0.95$; Depth $R^2 \approx 0.90$), confirming that interaction structure plays a central role in cascade geometry. The majority/minority framework gives slightly lower absolute error but slightly worse pseudo-$R^2$ and RMSE, so neither encoding dominates. Its residual $\log(\text{size}) \times$ platform interaction is smaller in the composition-only models and larger once alignment is included, which is expected: recoding composition by each platform's dominant ideology absorbs part of the platform contrast directly into the composition terms, and that absorption is redundant once alignment is in the model. The attenuation conclusion is the same under both encodings. We retain the left/right specification as our main model because its coefficients carry a fixed ideological meaning across platforms.

## M Marginal Effects of Ideological Structure

Figure 12 plots marginal effects of alignment ratio, left-user ratio, and right-user ratio on cascade structure. For **depth**, higher alignment is associated with shallower structures, consistent with the idea that ideologically homogeneous cascades sustain fewer extended exchanges. Cascades with fewer right-leaning participants exhibit greater layering, echoing the motif finding that right-leaning users rarely construct fully right-aligned chains. The marginal effects of left- and right-user ratios are asymmetric: increases in right-leaning participation are linked to sharper declines in depth than comparable increases in left-leaning participation. For **breadth**, higher alignment corresponds to wider cascades, indicating that homogeneous groups are more likely to generate simultaneous uptake at the same level. The left-user ratio exhibits an inverted-U relationship with breadth, while the right-user ratio shows a more linear pattern: as right-leaning users become increasingly dominant, cascades grow consistently broader.

## N Platform Affordances and Technical Comparability

Our comparison rests on a narrow notion of technical comparability: we do not assume the platforms are identical exposure environments, but that they share Twitter-like affordances (posting, reposting, threaded replies) that enable construction of comparable cascade objects. Ranking systems, moderation decisions, and feed settings may affect which posts users see and which cascades grow; these are partly opaque and cannot be fully identified from public traces. Our analysis should be interpreted as an observational comparison of public interaction traces, not a causal estimate of platform design effects.

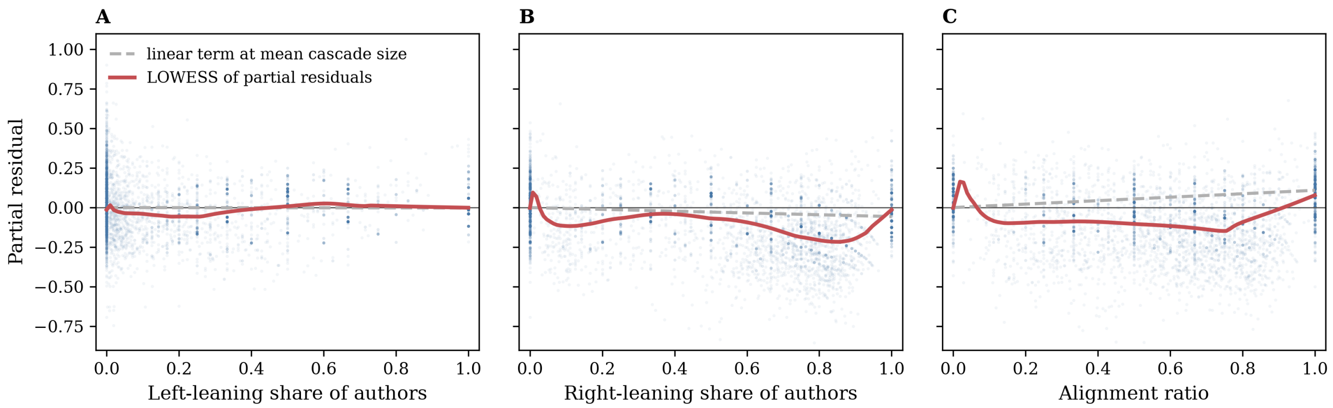


Figure 11: Partial residual plots under a linear baseline for (A) `author_left_ratio`, (B) `author_right_ratio`, and (C) `author_alignment_ratio`. Points are a random 20,000 of the 123,189 cascades; the fit uses all of them. The dashed line is the predictor's linear contribution evaluated at mean cascade size, the solid curve a LOWESS smoother of the partial residuals. Separation between the two is the nonlinearity a spline basis would absorb.

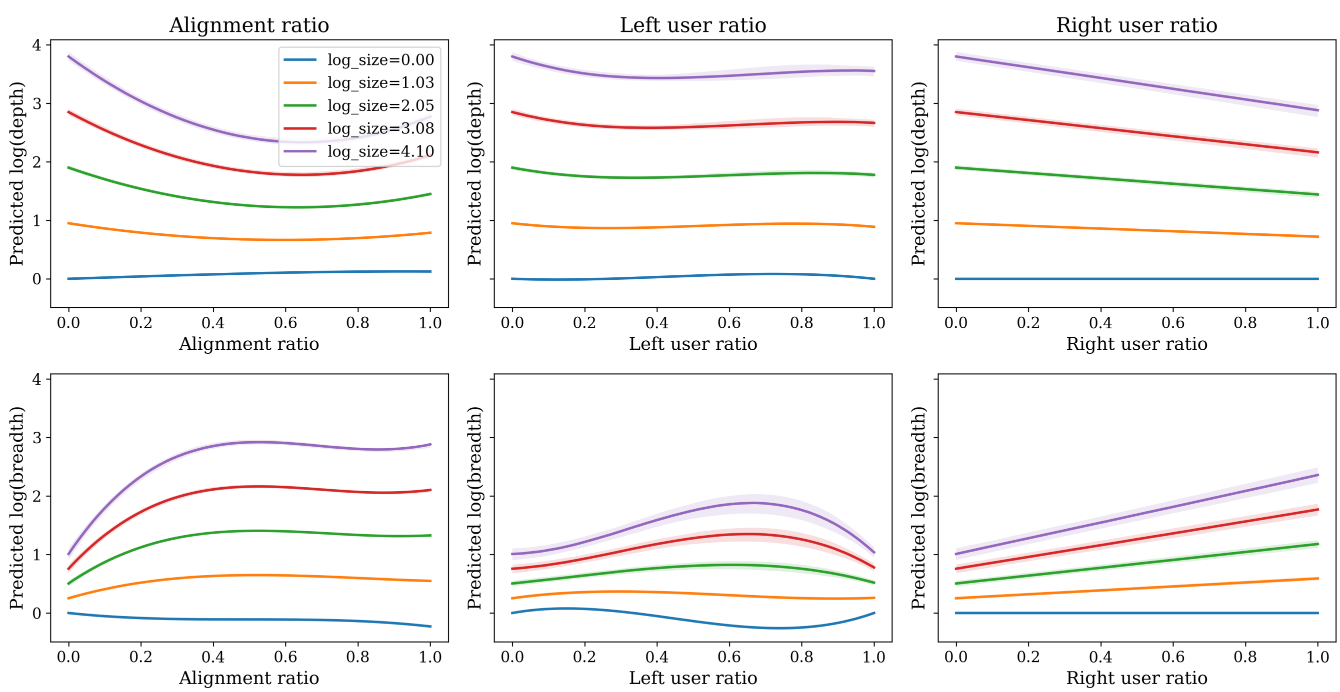


Figure 12: Marginal effects of ideological alignment, left-user ratio, and right-user ratio on cascade structure. Solid lines denote predicted log(depth) (top row) and log(breadth) (bottom row) at five equally spaced values of log(size). Shaded bands represent 95% bootstrap confidence intervals. Homogeneous cascades (high alignment) tend to flatten into wide, shallow structures; heterogeneous cascades deepen through sequential exchanges.

Table 7: Comparison of user-composition encodings (Left/Right vs. Majority/Minority), with and without alignment. Robust Linear Models.

| | Panel A: Composition only | | | | Panel B: + Alignment (df=3) | | | |
|---|---|---|---|---|---|---|---|---|
| | Breadth (log_breadth) | | Depth (log_depth) | | Breadth (log_breadth) | | Depth (log_depth) | |
| | L/R | Maj/Min | L/R | Maj/Min | L/R+Align | Maj/Min+Align | L/R+Align | Maj/Min+Align |
| Observations | 120,693 | 120,693 | 120,693 | 120,693 | 120,693 | 120,693 | 120,693 | 120,693 |
| Pseudo $R^2$ | **0.9367** | 0.9359 | **0.8861** | 0.8825 | **0.9548** | 0.9526 | **0.9078** | 0.9064 |
| MAE | 0.0645 | **0.0638** | 0.0500 | **0.0480** | 0.0468 | **0.0437** | 0.0402 | **0.0380** |
| RMSE | **0.1164** | 0.1172 | **0.0935** | 0.0949 | **0.0984** | 0.1008 | **0.0841** | 0.0848 |
| $\beta_{\log(\text{size})}$ | 0.3718 | 0.3552 | 0.8411 | 0.8285 | 0.2134 | 0.4106 | 0.9478 | 0.8093 |
| $\beta_{\log(\text{size})\times\text{platform(ts)}}$ | $0.2847$ | $0.1250$ | $-0.1913$ | $-0.1093$ | $0.0662$ | $0.1216$ | $-0.0455$ | $-0.0970$ |

*Notes:* L/R models the *left-user share* with a spline (df=3) and the *right-user share* linearly, interacted with log(size) and platform. Maj/Min re-codes composition by platform-dominant ideology (majority) vs. opposing (minority); majority share modeled with a spline (df=3), minority share linearly. Panel B additionally includes the *alignment ratio* (spline, df=3) and its interactions. All eight models are fitted with the Huber loss and Huber's proposal 2 scale, matching Table 1. The sample is the 120,693 cascades for which reply-tree ideology shares could be recomputed, so $\beta_3$ differs slightly in level from Table 1. Bold marks the better of the two encodings within each column pair. Coefficients are reported without standard errors because only the comparison between encodings is at issue and both share the estimator.

The technical lineage of both platforms provides additional justification. Bluesky is built on the AT Protocol, which also supports custom feed generation through a feed-generator architecture. Truth Social has been widely documented as Mastodon-derived, using open-source social networking software with visual modifications. These lineages support structural comparability of cascade objects while reinforcing that ranking, moderation, and exposure should be treated as platform-specific and only partially observed.

Table 8: Platform affordances and technical lineage relevant to cascade construction.

| Dimension | Truth Social | Bluesky |
|---|---|---|
| Technical lineage | Mastodon-derived platform | Built on the open-source AT Protocol ecosystem |
| Posting affordance | Supports user-authored posts | Supports user-authored posts |
| Reposting affordance | Supports repost-style resharing | Supports repost-style resharing |
| Reply affordance | Supports threaded replies | Supports threaded replies |
| Cascade observability | Public repost and reply traces observable | Public repost and reply traces observable |
| Feed and ranking | Platform-specific visibility and ranking; not directly observed | Multiple feeds via feed-generator architecture; visibility varies |
| Moderation | Platform-specific moderation may affect observed patterns | Decentralized moderation and curation may affect patterns |
| Analytic implication | Enables cascade construction; exposure process not fully identified | Enables cascade construction; exposure process not fully identified |

# O Computing Environment

All experiments were conducted on a single-node, multi-GPU system equipped with **8 NVIDIA H100 GPUs (80GB memory each)**. Model inference was deployed using **vLLM** with **bfloat16 (BF16) precision**, enabling efficient batched inference while maintaining numerical stability for long-context inputs.